

“个体多样性自适应的多模态多目标差分进化算法”

补充材料

1 详细实验结果——对比实验

1.1 CEC 2019 MMOPs上各算法的实验结果

表 S1 MMODE-AID 与 7 个对比算法在 CEC 2019 MMOPs 上获得的 IGDX 均值和标准差

测试问题	DN- NSGAI1	MMEA-WI	MMOE-A- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MMF1	1.4011e-1 (2.11e-2) -	4.5661e-2 (2.91e-3) -	4.2639e-2 (2.60e-3) -	6.5482e-2 (7.08e-3) -	4.3044e-2 (3.05e-3) -	4.9820e-2 (1.28e-3) -	4.9673e-2 (1.50e-3) -	3.8675e-2 (1.83e-3)
MMF2	1.0494e-1 (2.32e-2) -	3.5249e-2 (1.44e-2) -	4.5510e-2 (1.19e-2) -	8.1122e-2 (3.46e-2) -	2.7673e-2 (1.16e-2) +	3.1737e-2 (7.97e-3) -	3.3449e-2 (5.14e-3) -	2.9114e-2 (9.25e-3)
MMF3	9.3320e-2 (2.44e-2) -	3.1960e-2 (9.77e-3) -	3.7185e-2 (7.55e-3) -	7.7047e-2 (3.31e-2) -	2.2843e-2 (9.49e-3) +	2.5008e-2 (5.62e-3) -	2.4008e-2 (5.16e-3) =	2.4049e-2 (1.53e-3)
MMF4	9.6121e-2 (1.71e-2) -	2.4919e-2 (1.49e-3) -	2.0368e-2 (1.04e-3) =	4.1537e-2 (5.40e-3) -	2.1164e-2 (1.17e-3) -	2.4897e-2 (7.17e-4) -	2.0296e-2 (1.13e-3) =	1.9501e-2 (9.79e-4)
MMF5	1.9960e-1 (2.12e-2) -	7.5971e-2 (2.73e-3) -	7.5566e-2 (5.37e-3) -	1.0446e-1 (8.00e-3) -	7.0598e-2 (3.39e-3) -	7.0382e-2 (5.02e-3) -	6.8949e-2 (3.33e-3) -	5.8214e-2 (1.70e-3)
MMF6	1.6267e-1 (1.59e-2) -	6.5061e-2 (2.64e-3) -	6.2425e-2 (2.93e-3) -	9.2411e-2 (1.07e-2) -	5.9288e-2 (2.34e-3) -	5.7518e-2 (1.40e-3) -	5.7605e-2 (2.59e-3) -	4.5156e-2 (1.86e-3)
MMF7	7.2417e-2 (1.01e-2) -	2.4874e-2 (2.67e-3) -	2.1113e-2 (2.60e-3) -	3.6921e-2 (4.83e-3) -	2.0155e-2 (1.50e-3) -	2.2515e-2 (8.61e-4) -	2.0055e-2 (1.33e-3) -	1.7731e-2 (2.56e-3)
MMF8	2.2559e-1 (2.87e-2) -	7.9480e-2 (1.62e-2) -	9.0157e-2 (1.99e-2) -	1.1034e-1 (2.72e-2) -	6.2186e-2 (1.00e-2) -	5.9448e-2 (7.09e-3) -	5.4750e-2 (7.52e-3) -	4.3635e-2 (3.68e-3)
MMF9	2.4877e-2 (4.10e-3) -	6.2638e-3 (2.52e-4) -	4.5259e-3 (2.03e-4) -	1.4260e-2 (6.51e-3) -	4.7164e-3 (2.68e-4) -	7.3813e-3 (5.66e-4) -	6.9691e-3 (4.31e-4) -	3.0462e-3 (1.93e-4)
MMF10	9.6387e-2 (2.90e-2) -	1.3905e-2 (6.52e-2) -	2.2154e-3 (4.08e-3) -	2.1711e-2 (7.62e-3) -	2.5093e-2 (8.91e-2) -	6.8719e-3 (4.26e-4) -	1.5657e-3 (6.21e-5) +	1.5959e-3 (2.37e-4)
MMF11	2.5531e-2 (7.08e-3) -	4.3861e-3 (2.36e-4) -	3.3552e-3 (1.72e-4) -	5.6668e-3 (3.12e-4) -	3.4509e-3 (1.32e-4) -	7.7743e-3 (4.71e-4) -	4.3382e-3 (2.18e-4) -	2.6302e-3 (1.21e-4)
MMF12	1.6870e-2 (5.10e-3) -	1.7691e-3 (9.43e-5) -	1.2899e-3 (9.78e-5) -	5.1026e-3 (7.57e-4) -	1.7358e-3 (1.50e-4) -	3.1184e-3 (2.58e-4) -	1.7140e-3 (1.36e-4) -	1.0232e-3 (8.83e-5)
MMF13	9.0794e-2 (2.00e-2) -	2.0216e-2 (2.88e-3) -	1.8731e-2 (5.85e-4) -	2.9648e-2 (1.16e-3) -	2.1301e-2 (9.02e-4) -	3.9021e-2 (1.05e-3) -	2.5988e-2 (7.75e-4) -	1.0380e-2 (4.72e-4)
MMF14	8.3230e-2 (5.40e-3) -	4.2428e-2 (7.40e-4) -	3.7914e-2 (2.73e-4) -	6.1197e-2 (3.25e-3) -	3.8838e-2 (5.13e-4) -	4.2194e-2 (5.22e-4) -	3.9746e-2 (5.32e-4) -	2.2385e-2 (1.11e-3)
MMF15	8.3532e-2 (1.01e-2) -	3.1297e-2 (5.89e-4) -	2.8712e-2 (4.25e-4) -	4.2530e-2 (1.49e-3) -	3.0139e-2 (5.39e-4) -	4.2324e-2 (5.76e-4) -	2.9219e-2 (5.29e-4) -	2.7349e-2 (5.73e-4)
MMF1_z	4.5119e-2 (8.08e-3) -	3.3591e-2 (2.14e-3) -	3.4645e-2 (4.26e-3) -	4.8533e-2 (6.89e-3) -	3.0746e-2 (1.65e-3) -	3.0404e-2 (9.55e-4) -	2.7281e-2 (1.45e-3) -	2.0125e-2 (2.08e-3)
MMF1_e	1.2952e+0 (5.56e-1) -	7.2500e-1 (1.63e-1) -	1.0462e+0 (3.84e-1) -	8.2344e-1 (3.47e-1) -	5.3729e-1 (1.95e-1) -	2.6616e-1 (2.54e-2) -	3.1958e-1 (7.62e-2) -	1.0382e-1 (5.02e-3)
MMF14_a	1.1549e-1 (9.90e-3) -	5.7730e-2 (1.49e-3) -	4.7918e-2 (5.74e-4) +	6.8417e-2 (3.40e-3) -	4.8183e-2 (3.87e-4) +	5.3510e-2 (7.29e-4) -	5.2079e-2 (1.78e-3) =	5.1174e-2 (8.55e-4)
MMF15_a	1.0787e-1 (1.23e-2) -	3.8093e-2 (7.31e-4) -	3.2579e-2 (5.88e-4) -	4.8070e-2 (2.05e-3) -	3.3339e-2 (5.86e-4) -	4.8620e-2 (1.52e-3) -	3.3964e-2 (7.07e-4) -	2.1731e-2 (9.60e-4)
Omni_test	1.5920e-1 (6.57e-2) -	5.5336e-1 (1.59e-1) -	1.1583e-1 (3.54e-2) -	6.1923e-1 (1.55e-1) -	5.7900e-1 (1.90e-1) -	7.8703e-2 (1.59e-3) -	9.1174e-2 (2.02e-2) -	5.3635e-2 (7.09e-3)
SYM_RART_simple	5.0916e-1 (6.58e-1) -	5.8449e-2 (3.02e-3) -	4.4559e-2 (4.03e-3) -	2.7222e-1 (3.02e-1) -	6.0665e-2 (1.28e-2) -	6.9133e-2 (4.02e-3) -	4.1254e-2 (2.96e-3) =	4.0462e-2 (2.68e-4)
SYM_RART_rotate d	9.7049e-1 (8.15e-1) -	2.6119e-1 (4.45e-1) -	4.5484e-1 (5.71e-1) -	2.5881e-1 (2.45e-1) -	1.1530e-1 (1.88e-1) -	9.6836e-2 (4.21e-3) -	8.4783e-2 (7.61e-3) -	7.5093e-2 (8.91e-2)
+/-/=	0/22/0	0/22/0	1/20/1	0/22/0	3/19/0	0/22/0	1/17/4	

表 S2 MMODE-AID 与 7 个对比算法在 CEC 2019 MMOPs 上获得的 IGDF 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MMF1	3.9670e-2 (1.97e-2) -	3.8082e-3 (2.75e-4) -	3.2159e-3 (1.55e-4) -	4.4866e-3 (3.98e-4) -	3.7944e-3 (2.03e-4) -	3.2219e-3 (1.24e-4) -	2.9855e-3 (1.27e-4) -	2.6098e-3 (1.02e-4)
MMF2	1.6876e-1 (5.05e-2) -	1.3565e-2 (7.80e-3) -	1.8246e-2 (5.55e-3) -	4.3886e-2 (1.47e-2) -	7.0576e-2 (1.14e-3) -	1.7780e-2 (4.72e-3) -	1.1945e-2 (1.71e-3) -	1.0744e-2 (1.27e-3)
MMF3	1.4746e-1 (5.10e-2) -	1.1158e-2 (2.68e-3) -	1.3184e-2 (3.03e-3) -	3.7929e-2 (1.39e-2) -	9.4553e-3 (1.39e-3) -	1.3382e-2 (2.41e-3) -	1.0271e-2 (1.94e-3) -	7.8273e-3 (1.25e-3)
MMF4	2.1680e-2 (7.21e-3) -	3.7432e-3 (2.58e-4) -	3.2896e-3 (1.43e-4) -	3.7503e-3 (2.31e-4) -	3.4534e-3 (1.93e-4) -	3.8058e-3 (1.43e-4) -	3.1647e-3 (1.70e-4) -	2.4467e-3 (7.15e-5)
MMF5	4.0477e-2 (1.92e-2) -	3.5755e-3 (2.55e-4) -	3.2393e-3 (1.58e-4) -	3.9793e-3 (2.67e-4) -	3.6144e-3 (2.13e-4) -	3.6011e-3 (1.46e-4) -	3.2277e-3 (1.68e-4) -	2.6628e-3 (7.98e-5)
MMF6	2.5948e-2 (1.09e-2) -	3.4938e-3 (2.44e-4) -	3.1986e-3 (1.25e-4) -	3.8315e-3 (2.69e-4) -	3.4926e-3 (1.65e-4) -	3.4619e-3 (1.53e-4) -	3.1950e-3 (1.58e-4) -	2.5432e-3 (1.24e-4)
MMF7	1.4596e-2 (2.34e-3) -	3.9173e-3 (2.65e-4) -	1.8893e-3 (1.46e-4) +	3.7897e-3 (3.66e-4) -	3.4440e-3 (1.33e-4) -	3.3405e-3 (1.20e-4) -	3.0323e-3 (1.15e-4) -	2.1255e-3 (4.61e-5)
MMF8	1.8065e-2 (4.24e-3) -	3.6830e-3 (1.97e-4) -	3.7445e-3 (2.08e-4) -	4.7112e-3 (2.10e-4) -	4.0850e-3 (2.48e-4) -	5.1861e-3 (4.01e-4) -	3.6065e-3 (2.01e-4) -	2.3116e-3 (2.15e-4)
MMF9	4.4290e-2 (7.10e-3) -	1.5641e-2 (1.20e-3) -	1.3753e-2 (1.02e-3) -	1.6996e-2 (1.60e-3) -	1.4132e-2 (1.16e-3) -	1.1529e-2 (1.29e-3) -	1.2225e-2 (6.39e-4) -	9.5940e-3 (4.22e-4)
MMF10	2.8837e-1 (3.71e-2) -	2.2994e-2 (5.24e-2) -	9.9863e-3 (7.50e-3) -	6.3879e-2 (1.18e-2) -	9.1091e-3 (9.40e-4) -	2.3733e-2 (1.25e-3) -	6.7021e-3 (3.67e-4) -	6.4927e-3 (5.37e-2)
MMF11	7.1899e-2 (1.39e-2) -	1.5620e-2 (1.97e-3) -	1.3529e-2 (1.06e-3) -	1.6571e-2 (1.58e-3) -	1.3361e-2 (1.21e-3) -	2.1185e-2 (8.40e-4) -	1.0139e-2 (4.67e-4) -	9.2348e-3 (4.05e-4)
MMF12	4.7366e-2 (1.42e-2) -	2.8630e-3 (2.02e-4) -	2.4029e-3 (1.04e-4) -	1.2587e-2 (2.26e-3) -	2.7174e-3 (1.59e-4) -	5.1043e-3 (2.75e-4) -	2.0267e-3 (9.10e-5) =	2.0211e-3 (7.54e-5)
MMF13	8.8169e-2 (1.95e-2) -	1.6215e-2 (1.57e-3) -	1.5104e-2 (2.04e-3) -	1.3172e-2 (6.12e-4) -	1.3858e-2 (9.73e-4) -	2.6472e-2 (8.02e-4) -	1.1316e-2 (4.96e-4) -	7.8246e-3 (1.09e-4)
MMF14	1.3300e-1 (9.95e-3) -	6.9447e-2 (1.16e-3) -	6.4308e-2 (1.00e-3) -	8.1788e-2 (2.47e-3) -	6.5502e-2 (9.83e-4) -	6.1852e-2 (9.07e-4) -	6.2488e-2 (7.47e-4) -	5.7218e-2 (1.38e-3)
MMF15	1.8803e-1 (2.08e-2) -	6.7836e-2 (1.63e-3) -	6.0829e-2 (7.20e-4) -	8.3776e-2 (2.33e-3) -	6.2241e-2 (7.73e-4) -	8.7363e-2 (1.25e-3) -	5.9545e-2 (8.57e-4) =	5.7363e-2 (1.52e-3)
MMF1_z	2.9217e-2 (1.04e-2) -	3.7599e-3 (2.31e-4) -	3.1264e-3 (1.71e-4) -	4.2672e-3 (3.10e-4) -	3.6414e-3 (1.87e-4) -	3.3336e-3 (1.49e-4) -	2.8929e-3 (1.24e-4) -	2.5250e-3 (1.43e-4)
MMF1_e	1.3683e-1 (7.23e-2) -	3.0328e-2 (1.69e-2) -	5.3384e-3 (1.46e-3) -	1.1373e-2 (1.49e-3) -	1.0495e-2 (1.46e-3) -	1.0634e-2 (3.58e-3) -	1.0738e-2 (1.61e-3) -	4.1495e-3 (1.52e-4)
MMF14_a	1.5964e-1 (1.40e-2) -	7.4047e-2 (1.77e-3) -	6.5888e-2 (1.04e-3) -	8.0618e-2 (2.37e-3) -	6.8056e-2 (9.63e-4) -	6.4575e-2 (1.19e-3) -	6.5487e-2 (9.09e-4) -	6.0262e-3 (6.96e-4)
MMF15_a	2.1147e-1 (2.03e-2) -	7.1379e-2 (1.46e-3) -	6.3126e-2 (9.20e-4) -	8.4908e-2 (2.61e-3) -	6.5215e-2 (9.17e-4) -	8.7601e-2 (2.65e-3) -	6.2200e-2 (1.24e-3) =	6.1299e-2 (8.47e-4)
Omni_test	1.0203e-1 (1.71e-2) -	1.2989e-2 (2.72e-3) -	1.1181e-2 (2.56e-3) =	4.2336e-2 (6.09e-3) -	2.3812e-2 (6.67e-3) -	1.8005e-2 (1.85e-3) -	1.1338e-2 (2.69e-3) -	1.0841e-2 (1.99e-3)
SYM_RART_simple	6.2811e-2 (9.74e-3) -	1.5941e-2 (1.94e-3) -	1.4483e-2 (1.29e-3) -	3.6033e-2 (4.33e-3) -	1.6458e-2 (2.09e-3) -	1.7518e-2 (2.00e-3) -	1.4254e-2 (1.61e-3) -	1.3298e-2 (1.41e-3)
SYM_RART_rotate d	9.8232e-2 (1.66e-2) -	1.8591e-2 (1.69e-3) -	1.6273e-2 (1.86e-3) -	4.4384e-2 (6.38e-3) -	1.7679e-2 (2.17e-3) -	2.6002e-2 (2.32e-3) -	1.7073e-2 (1.92e-3) -	1.5534e-2 (2.84e-3)
+/-/=	0/22/0	0/22/0	1/20/1	0/22/0	0/22/0	0/22/0	0/19/3	

表 S3 MMODE-AID 与 7 个对比算法在 CEC 2019 MMOPs 上获得的 PSP 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MMF1	6.8404e+0 (1.23e+0) -	1.8920e+1 (1.44e+0) -	2.3367e+1 (1.37e+0) -	1.5242e+1 (1.76e+0) -	2.2239e+1 (1.46e+0) -	2.0014e+1 (7.99e-1) -	2.0121e+1 (9.23e-1) -	2.5848e+1 (1.21e-1)
MMF2	9.3038e+0 (2.13e+0) -	2.8355e+1 (8.67e+0) -	2.1504e+1 (6.42e+0) -	1.2298e+1 (4.13e+0) -	3.4340e+1 (1.50e+1) =	3.1494e+1 (7.39e+0) -	2.9882e+1 (6.89e+0) -	3.4346e+1 (8.19e+0)
MMF3	1.0549e+1 (3.16e+0) -	3.0456e+1 (9.07e+0) -	2.6780e+1 (5.59e+0) -	1.2594e+1 (6.03e+0) -	4.1568e+1 (1.38e+1) =	3.9881e+1 (8.55e+0) -	4.1570e+1 (8.22e+0) =	4.1577e+1 (1.18e+0)
MMF4	1.0336e+1 (1.79e+0) -	4.0118e+1 (2.34e+0) -	4.9046e+1 (2.54e+0) -	2.4066e+1 (3.07e+0) -	4.7133e+1 (2.84e+0) -	4.0161e+1 (1.17e+0) -	4.9270e+1 (2.70e+0) -	5.1245e+1 (3.27e-1)
MMF5	4.6791e+0 (7.06e-1) -	1.3135e+1 (4.78e-1) -	1.3231e+1 (9.45e-1) -	9.5291e+0 (7.70e-1) -	1.3157e+1 (6.58e-1) -	1.4124e+1 (4.00e-1) -	1.4425e+1 (6.98e-1) -	1.7174e+1 (7.71e-1)
MMF6	5.6578e+0 (6.76e-1) -	1.5338e+1 (6.26e-1) -	1.6005e+1 (7.56e-1) -	1.0816e+1 (1.19e+0) -	1.6847e+1 (7.01e-1) -	1.6382e+1 (4.27e-1) -	1.6973e+1 (1.04e+0) -	2.2114e+1 (7.49e-1)
MMF7	1.3534e+1 (2.05e+0) -	4.0105e+1 (4.05e+0) -	4.6781e+1 (5.29e+0) -	2.7056e+1 (3.89e+0) -	4.8689e+1 (3.25e+0) -	4.4350e+1 (1.62e+0) -	4.2843e+1 (3.01e+0) -	5.6355e+1 (5.04e+0)
MMF8	4.4259e+0 (5.50e-1) -	1.2409e+1 (2.46e+0) -	1.1043e+1 (2.76e+0) -	9.0563e+0 (2.13e+0) -	1.6051e+1 (2.50e+0) -	1.6813e+1 (9.88e-1) -	1.8263e+1 (2.58e+0) -	2.2917e+1 (1.72e-1)
MMF9	4.0188e+1 (6.21e+0) -	1.5676e+2 (6.08e+0) -	2.1128e+2 (9.58e+0) -	7.0109e+1 (1.80e+1) -	2.0231e+2 (8.71e+0) -	1.3515e+2 (9.13e+0) -	1.4103e+2 (9.01e+0) -	3.2827e+2 (7.37e+0)

表 S3 MMODE-AID 与 7 个对比算法在 CEC 2019 MMOPs 上获得的 PSP 均值和标准差 (续)

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MMF10	1.0258e+1 (3.50e+0) -	6.9578e+1 (9.53e+1) -	4.2418e+2 (1.64e+2) -	4.6042e+1 (1.68e+1) -	3.9851e+1 (8.07e+0) -	1.4506e+2 (9.07e+0) -	6.2658e+2 (2.80e+1) =	6.2660e+2 (1.52e+0)
MMF11	3.8399e+1 (9.83e+0) -	2.2339e+2 (1.21e+1) -	2.9803e+2 (1.58e+1) -	1.7606e+2 (1.01e+1) -	2.4005e+2 (1.11e+1) -	1.2810e+2 (8.05e+0) -	2.3005e+2 (1.16e+1) -	3.8019e+2 (9.34e+0)
MMF12	5.8557e+1 (1.61e+1) -	5.5686e+2 (3.10e+1) -	6.7965e+2 (6.18e+1) -	1.9592e+2 (3.07e+1) -	5.7607e+2 (5.64e+1) -	3.2041e+2 (2.72e+1) -	5.8285e+2 (4.50e+1) -	9.7732e+2 (3.53e+1)
MMF13	1.1011e+1 (2.48e+0) -	4.7414e+1 (1.21e+0) -	5.1370e+1 (1.70e+0) -	3.3502e+1 (1.34e+0) -	4.6911e+1 (1.90e+0) -	2.5623e+1 (7.32e-1) -	3.8396e+1 (1.11e+0) -	9.6339e+1 (3.28e+0)
MMF14	1.2012e+1 (7.59e-1) -	2.3176e+1 (4.11e-1) -	2.6335e+1 (1.90e-1) -	1.6283e+1 (8.64e-1) -	2.5732e+1 (3.37e-1) -	2.3603e+1 (2.95e-1) -	2.5064e+1 (3.36e-1) -	4.4672e+1 (6.18e-1)
MMF15	1.1928e+1 (1.44e+0) -	3.1463e+1 (6.01e-1) -	3.3435e+1 (5.20e-1) -	2.3437e+1 (8.17e-1) -	3.3170e+1 (5.90e-1) -	2.3532e+1 (3.18e-1) -	3.4215e+1 (6.15e-1) -	3.6564e+1 (8.29e-1)
MMF1_z	2.1178e+1 (7.72e-1) -	2.9333e+1 (4.48e-1) -	2.8171e+1 (3.52e+0) -	1.8617e+1 (7.23e-1) -	3.2521e+1 (1.77e+0) -	3.1691e+1 (2.53e-1) -	3.5546e+1 (3.25e-1) -	4.9689e+1 (2.69e-1)
MMF1_e	6.3596e-1 (1.04e+0) -	1.3261e+0 (5.02e-1) -	9.3823e-1 (4.27e-1) -	1.0828e+0 (8.53e-1) -	1.5906e+0 (7.16e-1) -	3.6588e+0 (6.69e-1) -	3.0455e+0 (6.12e-1) -	9.6320e+0 (7.84e-1)
MMF14_a	7.9373e+0 (1.74e+0) -	1.6714e+1 (1.90e+0) -	2.0862e+1 (2.51e-1) +	1.4563e+1 (2.74e+0) -	1.9556e+1 (1.66e-1) =	1.8592e+1 (1.06e+0) -	1.9201e+1 (1.89e+0) -	1.9532e+1 (6.07e-1)
MMF15_a	9.0560e+0 (5.49e-1) -	2.5299e+1 (3.92e-1) -	2.9704e+1 (5.50e-1) -	2.0102e+1 (5.47e-1) -	2.9004e+1 (5.27e-1) -	2.0158e+1 (3.91e-1) -	2.8771e+1 (6.70e-1) -	4.6017e+1 (7.45e-1)
Omni_test	6.2012e+0 (3.89e-1) -	1.8047e+0 (5.82e-1) -	8.1506e+0 (2.12e+0) -	1.5024e+0 (5.63e-1) -	1.6037e+0 (7.03e-1) -	1.1810e+1 (2.54e-1) -	1.0296e+1 (1.82e+0) -	1.8642e+1 (8.47e-1)
SYM_RART_simple	1.9036e+0 (7.49e-1) -	1.6930e+1 (8.71e-1) -	2.2308e+1 (2.07e+0) -	3.5539e+0 (1.71e+0) -	1.6285e+1 (2.67e+0) -	1.4461e+1 (8.37e-1) -	2.4139e+1 (1.78e+0) -	2.4714e+1 (1.13e-1)
SYM_RART_rotate d	1.0119e+0 (3.29e-1) -	3.8084e+0 (3.56e+0) -	2.1688e+0 (4.15e+0) -	3.8500e+0 (1.27e+0) -	8.4448e+0 (3.35e+0) -	1.0322e+1 (4.56e-1) -	1.1769e+1 (1.05e+0) -	1.3313e+1 (1.46e-1)
+/-/=	0/22/0	0/22/0	1/21/0	0/22/0	0/19/3	0/22/0	0/20/2	

1.2 MPolygon上各算法的实验结果

表 S4 MMODE-AID 与 7 个对比算法在 MPolygon 上获得的 IGDX 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MPolygon1	1.3281e-1 (1.23e-2) -	9.6829e-2 (2.75e-3) -	8.0304e-2 (1.21e-3) -	1.0984e-1 (5.40e-3) -	9.8342e-2 (2.09e-3) -	9.7043e-2 (1.99e-3) -	7.9971e-2 (1.16e-3) -	7.7373e-2 (9.87e-4)
MPolygon2	2.3903e-1 (3.60e-2) -	1.7734e-1 (4.92e-3) -	1.5561e-1 (3.48e-3) -	2.0672e-1 (1.14e-2) -	2.1288e-1 (9.16e-3) -	1.8756e-1 (6.46e-3) -	1.5764e-1 (4.99e-3) -	1.3698e-1 (1.92e-3)
MPolygon3	1.2392e-1 (7.68e-3) -	9.1400e-2 (2.18e-3) -	7.5983e-2 (7.27e-4) -	1.1067e-1 (3.15e-3) -	9.1041e-2 (1.26e-3) -	8.8168e-2 (1.28e-3) -	8.8051e-2 (2.75e-3) -	6.9354e-2 (7.87e-4)
MPolygon4	2.2261e-1 (1.77e-2) -	1.8208e-1 (4.47e-3) -	1.5955e-1 (2.87e-3) -	1.9232e-1 (5.88e-3) -	2.3286e-1 (1.30e-2) -	1.7993e-1 (3.77e-3) -	1.6145e-1 (4.57e-3) -	1.4987e-1 (3.24e-3)
MPolygon5	1.3261e-1 (9.29e-3) -	9.7452e-2 (1.87e-3) -	8.1634e-2 (7.02e-4) -	1.0958e-1 (3.00e-3) -	9.6944e-2 (1.14e-3) -	9.4192e-2 (1.07e-3) -	8.1460e-2 (7.51e-4) -	5.3772e-2 (1.76e-3)
MPolygon6	2.2412e-1 (2.20e-2) -	1.8021e-1 (4.91e-3) -	1.6039e-1 (2.55e-3) -	1.9124e-1 (6.30e-3) -	2.2797e-1 (1.08e-2) -	1.7871e-1 (4.08e-3) -	1.6215e-1 (3.65e-3) -	1.1795e-1 (1.44e-3)
MPolygon7	1.1406e-1 (6.28e-3) -	8.5391e-2 (1.29e-3) -	7.2021e-2 (5.71e-4) -	1.0038e-1 (2.83e-3) -	8.4655e-2 (6.82e-4) -	8.1557e-2 (8.32e-4) -	7.1500e-2 (7.66e-4) -	5.3042e-2 (1.18e-3)
MPolygon8	2.0198e-1 (2.27e-2) -	1.6271e-1 (4.06e-3) -	1.4265e-1 (2.32e-3) -	1.7274e-1 (4.12e-3) -	2.1390e-1 (7.69e-3) -	1.5786e-1 (3.11e-3) -	1.4579e-1 (2.74e-3) -	9.9456e-2 (1.11e-3)
MPolygon9	1.1647e-1 (7.76e-3) -	8.7499e-2 (1.31e-3) -	7.3734e-2 (4.99e-4) -	9.8407e-2 (2.58e-3) -	8.6905e-2 (7.37e-4) -	8.3304e-2 (8.47e-4) -	7.3041e-2 (4.85e-4) -	6.2695e-2 (1.41e-3)
MPolygon10	1.9935e-1 (1.56e-2) -	1.6490e-1 (2.73e-3) -	1.4598e-1 (3.09e-3) -	1.7432e-1 (4.44e-3) -	2.1830e-1 (1.29e-2) -	1.6249e-1 (2.89e-3) -	1.3637e-1 (2.97e-3) =	1.3626e-1 (1.36e-3)
MPolygon11	1.1929e-1 (5.68e-3) -	8.8478e-2 (1.76e-3) -	7.4741e-2 (5.74e-4) -	9.9386e-2 (2.56e-3) -	8.8130e-2 (1.69e-3) -	8.4366e-2 (5.57e-4) -	7.3958e-2 (4.56e-4) -	4.0653e-2 (4.53e-4)
MPolygon12	2.0478e-1 (1.77e-2) -	1.6757e-1 (4.17e-3) -	1.4913e-1 (2.24e-3) -	1.7447e-1 (5.15e-3) -	2.1954e-1 (1.06e-2) -	1.6145e-1 (3.41e-3) -	1.4803e-1 (3.23e-3) -	1.2450e-1 (1.39e-3)
+/-/=	0/12/0	0/12/0	0/12/0	0/12/0	0/12/0	0/12/0	0/11/1	

表 S5 MMODE-AID 与 7 个对比算法在 MPolygon 上获得的 IGDF 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MPolygon1	7.0907e-2 (5.13e-3) -	6.1180e-2 (1.52e-3) -	5.4820e-2 (1.29e-3) -	4.9958e-2 (1.01e-3) +	6.6895e-2 (1.98e-3) -	7.5663e-2 (1.13e-3) -	7.4929e-2 (1.01e-3) -	5.0454e-2 (1.79e-3)

表 S5 MMODE-AID 与 7 个对比算法在 MPolygon 上获得的 IGDF 均值和标准差 (续)

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MPolygon2	1.0702e-1 (2.35e-2) -	7.7658e-2 (2.51e-3) -	7.0860e-2 (1.45e-3) -	8.5684e-2 (3.85e-3) -	9.1849e-2 (3.90e-3) -	8.0936e-2 (2.48e-3) -	7.0936e-2 (1.79e-3) -	5.5454e-2 (1.15e-3)
MPolygon3	7.8492e-2 (3.61e-3) -	6.8120e-2 (1.34e-3) -	6.1304e-2 (7.77e-4) -	7.5893e-2 (2.40e-3) -	7.3797e-2 (1.30e-3) -	5.9771e-2 (6.56e-4) -	6.1273e-2 (8.51e-4) -	4.7902e-2 (5.30e-4)
MPolygon4	1.3743e-1 (1.62e-2) -	1.0716e-1 (2.23e-3) -	9.8765e-2 (2.11e-3) -	1.1068e-1 (2.80e-3) -	1.3261e-1 (5.51e-3) -	1.0539e-1 (1.63e-3) -	9.8367e-2 (1.93e-3) -	5.6773e-2 (7.08e-4)
MPolygon5	9.5490e-2 (4.98e-3) -	8.2220e-2 (1.36e-3) -	7.4236e-2 (1.13e-3) -	8.4795e-2 (2.38e-3) -	8.9056e-2 (1.40e-3) -	7.2055e-2 (7.92e-4) -	7.4804e-2 (9.84e-4) -	5.2238e-2 (1.15e-3)
MPolygon6	1.3292e-1 (1.74e-2) -	1.0669e-1 (2.28e-3) -	9.8479e-2 (1.26e-3) -	1.1117e-1 (2.96e-3) -	1.3027e-1 (4.53e-3) -	1.0479e-1 (1.69e-3) -	9.8844e-2 (1.84e-3) -	5.6404e-2 (7.56e-4)
MPolygon7	9.1676e-2 (3.60e-3) -	7.9119e-2 (1.32e-3) -	7.1831e-2 (8.61e-4) -	8.5576e-2 (2.12e-3) -	8.5417e-2 (1.32e-3) -	6.8532e-2 (6.88e-4) -	7.2322e-2 (9.36e-4) -	3.6964e-2 (7.32e-4)
MPolygon8	1.3292e-1 (1.50e-2) -	1.0411e-1 (2.22e-3) -	9.5803e-2 (1.57e-3) -	1.0799e-1 (2.58e-3) -	1.2734e-1 (3.27e-3) -	9.9471e-2 (1.30e-3) -	9.4581e-2 (1.38e-3) -	4.9786e-2 (6.82e-4)
MPolygon9	1.0024e-1 (3.89e-3) -	8.7819e-2 (1.32e-3) -	7.9696e-2 (1.01e-3) =	9.0374e-2 (2.11e-3) -	9.5294e-2 (1.37e-3) -	7.9689e-2 (7.38e-4) =	8.0139e-2 (1.28e-3) -	7.9685e-2 (8.38e-4)
MPolygon10	1.4410e-1 (1.17e-2) -	1.1473e-1 (1.92e-3) -	1.0634e-1 (1.47e-3) -	1.1884e-1 (2.44e-3) -	1.4296e-1 (6.71e-3) -	1.1145e-1 (2.31e-3) -	1.0437e-1 (1.79e-3) -	6.5977e-2 (6.42e-4)
MPolygon11	1.1023e-1 (3.85e-3) -	9.4891e-2 (1.60e-3) -	8.6052e-2 (1.08e-3) -	9.8887e-2 (2.11e-3) -	1.0202e-1 (1.60e-3) -	8.2106e-2 (6.74e-4) -	8.6465e-2 (1.00e-3) -	4.0421e-2 (4.73e-4)
MPolygon12	1.5898e-1 (1.87e-2) -	1.2476e-1 (2.35e-3) -	1.1649e-1 (2.00e-3) -	1.2841e-1 (3.09e-3) -	1.5391e-1 (5.45e-3) -	1.1977e-1 (1.48e-3) -	1.1395e-1 (2.06e-3) -	4.2403e-2 (6.25e-4)
+/-/=	0/12/0	0/12/0	0/11/1	1/11/0	0/12/0	0/11/1	0/12/0	

表 S6 MMODE-AID 与 7 个对比算法在 MPolygon 上获得的 PSP 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MPolygon1	7.4482e+0 (6.50e-1) -	1.0229e+1 (2.92e-1) -	1.2451e+1 (1.83e-1) -	9.0157e+0 (4.32e-1) -	1.0162e+1 (2.14e-1) -	1.0301e+1 (2.11e-1) -	1.2493e+1 (1.84e-1) -	1.2924e+1 (1.94e-1)
MPolygon2	3.8663e+0 (5.89e-1) -	5.6307e+0 (1.57e-1) -	6.4101e+0 (1.51e-1) -	4.8338e+0 (2.67e-1) -	4.6919e+0 (2.07e-1) -	5.3278e+0 (1.82e-1) -	6.3372e+0 (2.00e-1) -	7.2942e+0 (1.16e-1)
MPolygon3	7.9353e+0 (5.01e-1) -	1.0933e+1 (2.53e-1) -	1.3157e+1 (1.28e-1) -	9.0357e+0 (2.56e-1) -	1.0982e+1 (1.49e-1) -	1.1334e+1 (1.64e-1) -	1.1278e+1 (1.03e-1) -	1.4413e+1 (1.56e-1)
MPolygon4	4.2990e+0 (3.48e-1) -	5.4836e+0 (1.35e-1) -	6.2668e+0 (1.11e-1) -	5.1991e+0 (1.63e-1) -	4.2943e+0 (2.42e-1) -	5.5576e+0 (1.15e-1) -	6.1937e+0 (1.76e-1) -	6.6721e+0 (1.25e-1)
MPolygon5	7.4918e+0 (5.23e-1) -	1.0253e+1 (1.95e-1) -	1.2246e+1 (1.05e-1) -	9.1256e+0 (2.53e-1) -	1.0312e+1 (1.22e-1) -	1.0612e+1 (1.20e-1) -	1.2272e+1 (1.14e-1) -	1.8593e+1 (1.08e-1)
MPolygon6	4.2754e+0 (4.67e-1) -	5.5311e+0 (1.50e-1) -	6.2344e+0 (9.88e-2) -	5.2284e+0 (1.69e-1) -	4.3812e+0 (2.24e-1) -	5.5884e+0 (1.30e-1) -	6.1599e+0 (1.38e-1) -	8.4778e+0 (1.24e-1)
MPolygon7	8.7492e+0 (4.81e-1) -	1.1703e+1 (1.77e-1) -	1.3880e+1 (1.11e-1) -	9.9582e+0 (2.82e-1) -	1.1803e+1 (9.50e-2) -	1.2253e+1 (1.25e-1) -	1.3982e+1 (1.50e-1) -	1.8850e+1 (1.47e-1)
MPolygon8	4.8853e+0 (5.25e-1) -	6.0590e+0 (1.53e-1) -	7.0011e+0 (1.12e-1) -	5.7816e+0 (1.39e-1) -	4.6730e+0 (1.68e-1) -	6.3341e+0 (1.25e-1) -	6.8515e+0 (1.29e-1) -	1.0040e+1 (1.39e-1)
MPolygon9	8.5768e+0 (5.02e-1) -	1.1231e+1 (1.71e-1) -	1.3561e+1 (9.13e-2) -	1.0157e+1 (2.68e-1) -	1.1501e+1 (9.78e-2) -	1.2004e+1 (1.22e-1) -	1.3681e+1 (9.12e-2) -	1.5948e+1 (5.15e-2)
MPolygon10	4.9496e+0 (3.99e-1) -	5.9657e+0 (1.01e-1) -	6.8528e+0 (1.48e-1) -	5.7363e+0 (1.42e-1) -	4.5770e+0 (2.84e-1) -	6.1532e+0 (1.10e-1) -	7.3329e+0 (1.50e-1) =	7.3389e+0 (1.19e-1)
MPolygon11	8.3276e+0 (4.07e-1) -	1.1107e+1 (2.25e-1) -	1.3370e+1 (1.03e-1) -	1.0057e+1 (2.54e-1) -	1.1341e+1 (2.08e-1) -	1.1851e+1 (7.80e-2) -	1.3510e+1 (8.32e-2) -	2.4580e+1 (4.84e-2)
MPolygon12	4.7269e+0 (4.12e-1) -	5.8712e+0 (1.50e-1) -	6.7055e+0 (1.01e-1) -	5.7311e+0 (1.68e-1) -	4.5547e+0 (2.29e-1) -	6.1935e+0 (1.31e-1) -	6.7552e+0 (1.47e-1) -	8.0318e+0 (2.79e-1)
+/-/=	0/12/0	0/12/0	0/12/0	0/12/0	0/12/0	0/12/0	0/11/1	

1.3 HYL上各算法的实验结果

表 S7 MMODE-AID 与 7 个对比算法在 HYL 上获得的 IGDX 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
HYL1	3.7149e-1 (4.23e-2) -	2.1027e-1 (6.41e-3) -	1.8771e-1 (1.64e-3) -	2.4477e-1 (8.69e-3) -	1.8905e-1 (2.36e-3) -	1.9954e-1 (2.41e-3) -	2.0005e-1 (3.04e-3) -	1.8634e-1 (9.61e-4)
HYL2	7.0358e-1 (2.75e-2) -	6.3955e-1 (1.31e-2) -	5.6605e-1 (4.43e-3) -	9.5182e-1 (1.53e-1) -	5.7488e-1 (8.22e-3) -	5.8438e-1 (9.35e-3) -	5.7736e-1 (5.10e-3) -	5.5735e-1 (4.35e-3)
HYL3	9.2033e-1 (1.93e-2) -	8.5886e-1 (1.39e-2) -	8.4497e-1 (1.57e-2) -	1.0685e+0 (1.38e-1) -	8.2856e-1 (9.86e-3) -	1.1731e+0 (3.53e-2) -	8.3938e-1 (1.01e-2) -	7.9303e-1 (8.18e-3)

表 S7 MMODE-AID 与 7 个对比算法在 HYL 上获得的 IGDX 均值和标准差 (续)

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
HYL4	1.9207e+0 (1.88e-1) -	1.0411e+0 (1.53e-2) -	9.9901e-1 (1.07e-2) -	1.8312e+0 (4.56e-1) -	1.0070e+0 (9.17e-3) -	1.0671e+0 (1.13e-2) -	1.0109e+0 (9.52e-3) -	9.5609e-1 (6.21e-3)
HYL5	3.8397e-1 (3.87e-2) -	2.8020e-1 (5.99e-3) -	2.4805e-1 (2.77e-3) -	4.3024e-1 (4.61e-2) -	2.5615e-1 (2.51e-3) -	3.0583e-1 (5.28e-3) -	2.5688e-1 (3.20e-3) -	2.4612e-1 (2.04e-3)
+/-/=	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	

表 S8 MMODE-AID 与 7 个对比算法在 HYL 上获得的 IGDF 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
HYL1	4.2304e-2 (1.12e-2) -	3.1238e-2 (6.60e-3) -	3.8994e-2 (9.25e-3) -	2.8638e-2 (7.15e-3) -	3.3348e-2 (5.65e-3) -	2.8451e-2 (5.20e-3) -	2.9422e-2 (6.66e-3) -	1.6809e-2 (2.56e-3)
HYL2	4.2465e-2 (7.34e-3) -	3.0553e-2 (5.76e-3) -	4.0851e-2 (9.08e-3) -	2.5288e-2 (5.54e-3) -	3.2036e-2 (7.28e-3) -	1.6701e-2 (2.24e-3) -	3.0969e-2 (5.04e-3) -	1.4105e-2 (1.95e-3)
HYL3	3.1095e-1 (5.04e-2) -	1.8173e-1 (9.66e-3) -	2.2674e-1 (1.56e-2) -	1.6893e-1 (7.50e-3) -	1.9746e-1 (9.39e-3) -	2.6127e-1 (1.15e-2) -	2.0050e-1 (1.06e-2) -	1.3548e-1 (2.18e-3)
HYL4	2.8227e-1 (2.62e-2) -	1.9973e-1 (1.17e-2) -	2.2810e-1 (1.31e-2) -	2.0906e-1 (1.02e-2) -	2.3706e-1 (1.24e-2) -	1.8756e-1 (4.69e-3) -	2.2200e-1 (1.29e-2) -	1.4430e-1 (2.64e-3)
HYL5	9.5460e-2 (1.67e-2) -	4.6416e-2 (3.51e-3) -	4.7632e-2 (3.84e-3) -	6.3281e-2 (1.04e-2) -	4.4796e-2 (3.96e-3) -	4.2893e-2 (2.08e-3) -	4.3369e-2 (2.40e-3) -	2.9436e-2 (1.40e-3)
+/-/=	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	

表 S9 MMODE-AID 与 7 个对比算法在 HYL 上获得的 PSP 均值和标准差

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
HYL1	2.6671e+0 (4.19e-1) -	4.6296e+0 (1.83e-1) -	5.3263e+0 (4.71e-2) -	4.0629e+0 (1.44e-1) -	5.2087e+0 (7.41e-2) -	5.0053e+0 (2.12e-1) -	4.9823e+0 (8.19e-2) -	5.3665e+0 (2.77e-2)
HYL2	5.0036e-1 (1.44e-1) -	1.5613e+0 (3.20e-2) -	1.7655e+0 (1.36e-2) -	1.0279e+0 (1.68e-1) -	1.7353e+0 (2.35e-2) -	1.5067e+0 (5.11e-2) -	1.7012e+0 (1.51e-2) -	1.7942e+0 (1.37e-2)
HYL3	8.0231e-1 (1.23e-1) -	1.1592e+0 (1.98e-2) -	1.1821e+0 (2.23e-2) -	9.3038e-1 (1.07e-1) -	1.2063e+0 (1.46e-2) -	8.1518e-1 (2.44e-2) -	1.1905e+0 (1.45e-2) -	1.2608e+0 (1.34e-2)
HYL4	2.7457e-1 (4.33e-2) -	9.6051e-1 (1.42e-2) -	9.9511e-1 (1.07e-2) -	5.4115e-1 (1.10e-1) -	9.9311e-1 (9.03e-3) -	9.1677e-1 (1.60e-2) -	9.8910e-1 (9.32e-3) -	1.0456e+0 (6.80e-3)
HYL5	2.0676e+0 (4.20e-1) -	3.5638e+0 (7.38e-2) -	4.0295e+0 (4.43e-2) -	2.3228e+0 (2.57e-1) -	3.0603e+0 (3.27e-2) -	3.2545e+0 (1.45e-1) -	3.8922e+0 (4.87e-2) -	4.0628e+0 (3.62e-2)
+/-/=	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	

1.4 Friedman检验排名结果

表 S10 MMODE-AID 与 7 个对比算法在全部 3 个测试问题集上获得的 Friedman 平均排名

问题集	评价指标	DN- NSGAI	MMEA- WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_ SPDN	CEA-LES	CMMO	MMODE- AID
CEC 2019 MMOPs	IGDX	7.82	5.11	3.73	6.77	3.68	4.57	3.07	1.25
	IGDF	8	5.11	3.41	6.45	4.52	5.07	2.43	1
	PSP	7.77	5.18	3.70	6.77	3.61	4.5	3.07	1.39
MPolygon	IGDX	7.58	5.25	2.58	6.5	6.5	4.167	2.38	1.04
	IGDF	7.83	4.83	2.83	5.75	6.67	3.58	3.33	1.17
	PSP	7.67	5.33	2.54	6.5	6.42	4.04	2.5	1
HYL	IGDX	7.2	5.4	2.4	7.4	2.9	5.9	3.8	1
	IGDF	8	4	6.4	3.8	5.4	3	4.4	1
	PSP	8	5	2.4	6.8	3.4	5.6	3.8	1
总排名	IGDX	7.67	5.19	3.21	6.77	4.45	4.62	2.95	1.15
	IGDF	7.95	4.88	3.62	5.90	5.29	4.35	2.96	1.05
	PSP	7.77	5.23	3.18	6.72	4.37	4.53	2.99	1.22

2 详细实验结果——消融实验

2.1 CEC 2019 MMOPs上各算法的实验结果

表 S11 MMODE-AID 与 5 个变种算法在 CEC 2019 MMOPs 上获得的 IGD_X 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MMF1	8.0374e-2 (5.44e-3) -	5.7024e-2 (9.35e-4) -	4.9476e-2 (3.68e-3) -	3.8873e-2 (1.58e-3) =	4.1570e-2 (9.21e-3) -	3.8675e-2 (1.83e-3)
MMF2	6.9756e-2 (8.06e-3) -	4.9338e-2 (8.37e-3) -	4.1395e-2 (1.20e-2) -	5.7610e-2 (1.87e-2) -	1.1924e-1 (1.44e-1) -	2.9114e-2 (9.25e-3)
MMF3	5.1484e-2 (5.05e-3) -	4.3439e-2 (3.74e-3) -	3.4722e-2 (8.73e-3) -	4.1010e-2 (1.88e-2) -	2.8850e-2 (1.58e-2) -	2.4049e-2 (1.53e-3)
MMF4	8.8330e-2 (1.20e-2) -	4.1951e-2 (9.63e-4) -	3.8000e-2 (1.29e-3) -	1.9457e-2 (6.44e-4) =	2.0461e-2 (2.49e-3) =	1.9501e-2 (9.79e-4)
MMF5	1.1429e-1 (6.29e-3) -	6.2766e-2 (1.64e-3) -	5.1466e-2 (3.88e-3) +	6.6223e-2 (1.89e-3) -	7.3333e-2 (1.20e-2) -	5.8214e-2 (1.70e-3)
MMF6	1.0101e-1 (6.39e-3) -	5.5719e-2 (1.41e-3) -	5.9011e-2 (3.21e-3) -	5.7458e-2 (1.99e-3) -	5.9366e-2 (4.49e-3) -	4.5156e-2 (1.86e-3)
MMF7	9.6193e-2 (1.64e-2) -	3.1070e-2 (9.82e-4) -	2.2760e-2 (3.88e-3) -	2.1532e-2 (2.85e-3) -	2.2269e-2 (2.83e-3) -	1.7731e-2 (2.56e-3)
MMF8	3.1871e-1 (2.83e-2) -	6.4859e-2 (3.92e-3) -	5.1762e-2 (4.38e-3) -	5.6481e-2 (3.75e-3) -	5.4903e-2 (6.84e-3) -	4.3635e-2 (3.68e-3)
MMF9	4.8087e-2 (6.32e-3) -	1.2953e-2 (7.02e-4) -	6.1899e-3 (6.35e-4) -	5.5608e-3 (1.78e-4) -	5.7674e-3 (2.86e-4) -	3.0462e-3 (1.93e-4)
MMF10	5.0189e-2 (1.31e-2) -	2.5436e-3 (3.01e-4) -	2.5272e-2 (8.98e-2) -	3.8344e-2 (8.98e-2) -	1.6054e-3 (7.65e-2) =	1.5959e-3 (2.37e-4)
MMF11	6.4414e-2 (1.63e-2) -	8.7573e-3 (4.34e-4) -	4.2288e-3 (1.46e-4) -	4.1446e-3 (1.56e-4) -	4.1446e-3 (1.74e-4) -	2.6302e-3 (1.21e-4)
MMF12	4.4740e-2 (1.17e-2) -	2.9050e-3 (1.68e-4) -	1.6202e-3 (6.97e-5) -	1.5029e-3 (5.28e-5) -	1.5406e-3 (5.89e-5) -	1.0232e-3 (8.83e-5)
MMF13	1.8185e-1 (4.67e-2) -	3.1728e-2 (9.71e-4) -	2.3706e-2 (2.75e-3) -	1.9019e-2 (2.42e-4) -	2.5055e-2 (4.42e-3) -	1.0380e-2 (4.72e-4)
MMF14	1.4358e-1 (2.36e-2) -	6.8537e-2 (2.16e-3) -	4.3884e-2 (1.09e-3) -	4.2761e-2 (5.35e-4) -	4.4623e-2 (2.02e-3) -	2.2385e-2 (1.11e-3)
MMF15	1.6778e-1 (4.63e-2) -	5.6333e-2 (1.77e-3) -	3.3309e-2 (6.22e-4) -	3.3291e-2 (4.79e-4) -	3.3367e-2 (7.65e-4) -	2.7349e-2 (5.73e-4)
MMF1_z	1.7921e-1 (3.57e-2) -	7.4301e-2 (2.14e-3) -	5.2929e-2 (1.29e-3) -	3.0536e-2 (4.93e-3) -	3.1142e-2 (7.07e-3) -	2.0125e-2 (2.08e-3)
MMF1_e	9.0539e-1 (4.63e-2) -	6.3429e-1 (2.36e-3) -	1.0318e-1 (1.15e-3) =	2.9932e-1 (1.14e-1) -	2.4508e-1 (1.03e-1) -	1.0382e-1 (5.02e-3)
MMF14_a	2.3507e-1 (7.74e-3) -	9.0252e-2 (1.06e-3) -	8.8943e-2 (2.89e-3) -	5.1232e-2 (9.19e-4) =	5.4549e-2 (3.06e-3) -	5.1174e-2 (8.55e-4)
MMF15_a	4.0549e-1 (8.48e-2) -	6.2435e-1 (2.40e-1) -	6.3670e-1 (2.77e-1) -	3.3899e-2 (4.37e-4) -	3.6745e-2 (9.95e-4) -	2.1731e-2 (9.60e-4)
Omni_test	6.3897e-1 (3.51e-2) -	8.2658e-2 (7.11e-3) -	8.3934e-2 (1.32e-2) -	1.0135e-1 (5.31e-3) -	1.0051e-1 (2.84e-2) -	5.3635e-2 (7.09e-3)
SYM_RART_simple	1.8248e+0 (3.13e-1) -	7.2062e-2 (6.21e-3) -	5.7487e-2 (5.84e-3) -	1.0918e-1 (1.19e-2) -	1.2450e-1 (1.03e-2) -	4.0462e-2 (2.68e-4)
SYM_RART_rotated	1.7170e+0 (2.74e-1) -	9.0597e-2 (1.64e-2) -	7.5832e-2 (5.16e-3) =	1.3990e-1 (1.18e-2) -	1.2655e-1 (9.59e-3) -	7.5093e-2 (8.91e-2)
+/-/=	0/22/0	0/22/0	1/19/2	0/19/3	0/20/2	

表 S12 MMODE-AID 与 5 个变种算法在 CEC 2019 MMOPs 上获得的 IGDF 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MMF1	3.2225e-2 (4.77e-3) -	5.0387e-3 (1.70e-4) -	3.6405e-3 (1.04e-4) -	2.6234e-3 (1.05e-4) =	2.6202e-3 (1.67e-4) =	2.6098e-3 (1.02e-4)
MMF2	1.7046e-1 (5.03e-2) -	6.3037e-2 (1.10e-3) -	1.2178e-2 (2.92e-3) -	2.1930e-2 (5.09e-3) -	1.0155e-2 (1.25e-3) =	1.0744e-2 (1.27e-3)
MMF3	1.6056e-1 (4.19e-2) -	8.5561e-3 (6.86e-4) -	9.8022e-3 (1.15e-3) -	1.4546e-2 (6.36e-3) -	9.0006e-3 (1.34e-3) -	7.8273e-3 (1.25e-3)
MMF4	2.3166e-2 (4.83e-3) -	5.7787e-3 (2.62e-4) -	2.9883e-3 (1.22e-4) -	2.4581e-3 (1.25e-4) =	2.5859e-3 (1.79e-4) -	2.4467e-3 (7.15e-5)
MMF5	3.0809e-2 (6.04e-3) -	5.0450e-3 (2.28e-4) -	2.7279e-3 (1.18e-4) -	2.7214e-3 (9.27e-5) =	2.7054e-3 (1.48e-4) =	2.6628e-3 (7.98e-5)
MMF6	2.6964e-2 (4.28e-3) -	5.0015e-3 (2.25e-4) -	2.8902e-3 (9.16e-4) -	2.6170e-3 (1.82e-4) =	2.6230e-3 (1.36e-4) =	2.5432e-3 (1.24e-4)

表 S12 MMODE-AID 与 5 个变种算法在 CEC 2019 MMOPs 上获得的 IGDF 均值和标准差 (续)

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MMF7	3.1280e-2 (7.62e-3) -	5.9843e-3 (2.56e-4) -	2.4491e-3 (7.31e-5) -	2.3948e-3 (8.15e-5) -	2.4784e-3 (1.13e-4) -	2.1255e-3 (4.61e-5)
MMF8	6.0772e-2 (1.37e-2) -	5.4834e-3 (3.01e-4) -	3.2625e-3 (2.01e-4) -	3.6611e-3 (3.29e-4) -	3.6845e-3 (3.02e-4) -	2.3116e-3 (2.15e-4)
MMF9	9.5160e-2 (1.28e-2) -	1.9721e-2 (7.59e-4) -	9.5131e-3 (6.15e-4) =	9.5347e-3 (3.97e-4) =	9.5644e-3 (3.45e-4) =	9.5904e-3 (4.22e-4)
MMF10	2.9736e-1 (4.15e-2) -	1.6856e-2 (4.13e-3) -	2.5903e-2 (7.45e-2) -	7.3002e-2 (8.38e-2) -	2.0137e-2 (6.41e-2) -	6.4927e-3 (5.37e-2)
MMF11	1.4876e-1 (3.88e-2) -	1.9796e-2 (1.05e-3) -	9.9024e-3 (4.91e-4) -	9.4643e-3 (5.35e-4) -	9.3478e-3 (3.06e-4) -	9.2348e-3 (4.05e-4)
MMF12	1.3984e-1 (3.48e-2) -	3.4094e-3 (1.42e-4) -	2.0292e-3 (9.04e-5) =	1.9495e-3 (9.54e-5) =	1.9459e-3 (8.09e-5) =	2.0211e-3 (7.54e-5)
MMF13	4.4535e-1 (2.36e-1) -	1.7926e-2 (6.74e-4) -	8.8418e-3 (3.42e-4) -	8.7565e-3 (3.08e-4) -	8.7243e-3 (3.13e-4) -	7.8246e-3 (1.09e-4)
MMF14	3.0286e-1 (8.38e-2) -	1.0388e-1 (3.27e-3) -	6.8564e-2 (3.89e-3) -	5.8280e-2 (9.67e-4) =	5.8384e-2 (1.47e-3) =	5.7218e-2 (1.38e-3)
MMF15	3.8991e-1 (8.87e-2) -	1.1411e-1 (4.49e-3) -	6.9108e-2 (1.64e-3) -	5.9078e-2 (1.18e-3) -	5.9728e-2 (1.55e-3) -	5.7363e-2 (1.52e-3)
MMF1_z	3.7904e-1 (1.15e-1) -	1.0136e-1 (3.87e-3) -	5.9760e-2 (9.81e-4) -	2.5825e-3 (1.69e-4) =	2.6021e-3 (1.82e-4) =	2.5250e-3 (1.43e-4)
MMF1_e	4.4706e-1 (8.61e-2) -	1.2242e-1 (4.51e-3) -	6.1420e-2 (1.14e-3) -	8.3748e-3 (1.17e-3) -	8.2845e-3 (2.28e-3) -	4.1495e-3 (7.53e-4)
MMF14_a	3.1996e-2 (6.24e-3) -	9.1588e-2 (2.10e-3) -	8.5173e-2 (9.62e-3) -	6.0168e-2 (9.74e-4) -	6.0026e-2 (9.45e-4) -	6.0262e-3 (6.96e-4)
MMF15_a	2.4494e-1 (1.15e-1) -	8.1476e-2 (1.03e-3) -	7.1787e-2 (1.23e-3) -	6.0159e-2 (9.59e-4) +	6.1286e-2 (1.20e-3) =	6.1299e-2 (8.47e-4)
Omni_test	4.7878e-1 (1.14e-1) -	1.8894e-2 (2.11e-3) -	1.7370e-2 (2.27e-3) -	2.5783e-2 (4.14e-3) -	2.4343e-2 (3.35e-3) -	1.0841e-2 (1.99e-3)
SYM_RART_simple	1.1443e+0 (4.97e-1) -	3.4247e-2 (1.58e-3) -	1.5194e-2 (1.91e-3) -	2.6234e-3 (1.05e-4) -	3.1150e-2 (4.21e-3) -	1.3298e-2 (1.41e-3)
SYM_RART_rotated	1.0256e+0 (3.35e-1) -	1.6790e-2 (1.45e-3) -	1.6922e-2 (1.89e-3) -	2.1930e-2 (5.09e-3) -	3.4238e-2 (3.97e-3) -	1.5534e-2 (2.84e-3)
+/-/=	0/22/0	0/22/0	0/20/2	1/13/8	0/13/9	

表 S13 MMODE-AID 与 5 个变种算法在 CEC 2019 MMOPs 上获得的 PSP 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MMF1	1.2431e+1 (7.61e-1) -	1.7026e+1 (6.77e-1) -	2.0199e+1 (2.01e+0) -	2.5719e+1 (1.07e+0) =	2.4040e+1 (3.39e+0) -	2.5848e+1 (1.21e-1)
MMF2	1.4027e+1 (2.26e+0) -	2.0178e+1 (1.66e+1) -	2.3773e+1 (8.76e+0) -	1.7204e+1 (5.63e+0) -	8.2245e+0 (1.87e+1) -	3.4346e+1 (8.19e+0)
MMF3	1.8611e+1 (1.98e+0) -	2.2844e+1 (1.37e+1) -	2.8372e+1 (9.41e+0) -	2.3488e+1 (7.85e+0) -	3.3271e+1 (1.46e+1) -	4.1577e+1 (1.18e+0)
MMF4	1.1223e+1 (1.57e+0) -	2.3326e+1 (9.36e-1) -	2.6063e+1 (3.03e+0) -	5.1286e+1 (1.67e+0) =	4.8407e+1 (5.01e+0) -	5.1245e+1 (3.27e-1)
MMF5	8.7450e+0 (4.71e-1) -	1.1943e+1 (4.15e-1) -	1.9170e+1 (8.43e-1) +	1.5074e+1 (4.43e-1) -	1.3162e+1 (1.92e+0) -	1.7174e+1 (7.71e-1)
MMF6	9.8377e+0 (6.21e-1) -	1.5958e+1 (8.47e-1) -	1.6943e+1 (8.59e-1) -	1.7396e+1 (5.97e-1) -	1.6806e+1 (1.22e+0) -	2.2114e+1 (7.49e-1)
MMF7	1.0318e+1 (1.42e+0) -	3.2015e+1 (1.00e+0) -	4.3727e+1 (6.74e+0) -	4.6354e+1 (5.93e+0) -	4.4268e+1 (4.99e+0) -	5.6355e+1 (5.04e+0)
MMF8	3.1209e+0 (2.72e-1) -	1.5303e+1 (9.40e-1) -	1.6258e+1 (1.68e+0) -	1.7484e+1 (1.21e+0) -	1.8211e+1 (2.01e+0) -	2.2917e+1 (1.72e-1)
MMF9	2.0147e+1 (2.82e+0) -	7.7121e+1 (4.17e+0) -	1.5977e+2 (1.43e+1) -	1.7000e+2 (5.73e+0) -	1.7336e+2 (8.47e+0) -	3.2827e+2 (7.37e+0)
MMF10	1.9878e+1 (4.65e+0) -	3.9189e+2 (4.23e+1) -	3.7683e+1 (1.51e+2) -	2.5219e+1 (1.54e+2) -	6.2283e+2 (1.32e+2) =	6.2600e+2 (1.52e+0)
MMF11	1.5455e+1 (4.64e+0) -	1.1406e+2 (5.65e+0) -	2.3640e+2 (8.15e+0) -	2.5415e+2 (1.02e+1) -	2.4113e+2 (1.02e+1) -	3.8019e+2 (9.34e+0)
MMF12	2.2144e+1 (6.19e+0) -	3.4336e+2 (2.03e+1) -	6.1716e+2 (2.68e+1) -	6.6518e+2 (2.39e+1) -	6.4817e+2 (2.39e+1) -	9.7732e+2 (3.53e+1)
MMF13	5.4797e+0 (1.66e+0) -	3.1515e+1 (9.31e-1) -	4.2144e+1 (4.26e+0) -	5.2577e+1 (6.66e-1) -	3.9904e+1 (6.04e+0) -	9.6339e+1 (3.28e+0)
MMF14	6.0237e+0 (1.26e+0) -	1.4105e+1 (4.56e-1) -	2.2701e+1 (5.58e-1) -	2.1389e+1 (2.93e-1) -	2.2401e+1 (9.60e-1) -	4.4672e+1 (6.18e-1)
MMF15	5.5490e+0 (1.53e+0) -	1.7728e+1 (5.57e-1) -	3.0020e+1 (5.64e-1) -	3.0024e+1 (4.31e-1) -	2.9485e+1 (6.91e-1) -	3.6564e+1 (8.29e-1)

表 S13 MMODE-AID 与 5 个变种算法在 CEC 2019 MMOPs 上获得的 PSP 均值和标准差 (续)

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MMF1_z	5.4352e+0 (1.42e+0) -	1.3449e+1 (3.83e-1) -	1.8604e+1 (4.56e-1) -	3.2274e+1 (3.83e+0) -	3.2096e+1 (5.70e+0) -	4.9689e+1 (2.69e-1)
MMF1_e	8.7754e+0 (1.24e+0) -	1.5746e+0 (5.74e-1) -	9.0561e+0 (8.63e-1) -	3.3395e+0 (1.23e+0) -	3.9345e+0 (1.28e+0) -	9.6320e+0 (7.84e-1)
MMF14_a	4.1073e+0 (1.10e+0) -	1.0894e+1 (1.13e+0) -	1.1191e+1 (2.98e+0) -	1.9515e+1 (3.46e-1) =	1.8313e+1 (9.48e-1) -	1.9532e+1 (6.07e-1)
MMF15_a	2.1489e+0 (4.21e-1) -	1.5729e+0 (6.91e-1) -	1.5609e+0 (8.44e-1) -	2.3504e+1 (3.80e-1) -	2.7133e+1 (7.25e-1) -	4.6017e+1 (7.45e-1)
Omni_test	1.5397e+0 (8.80e-2) -	1.0159e+1 (1.06e+0) -	1.1094e+1 (1.41e+0) -	9.8125e+0 (5.21e-1) -	9.6413e+0 (1.82e+0) -	1.8642e+1 (8.47e-1)
SYM_RART_simple	5.4546e-1 (1.03e-1) -	1.3866e+1 (1.24e+0) -	1.7332e+1 (1.75e+0) -	9.1532e+0 (9.12e-1) -	8.0139e+0 (6.58e-1) -	2.4714e+1 (1.13e-1)
SYM_RART_rotated	5.8131e-1 (9.78e-2) -	1.0376e+1 (1.84e+0) -	1.2172e+1 (8.90e-1) -	7.1245e+0 (6.00e-1) -	7.6453e+0 (6.11e-1) -	1.3313e+1 (1.46e-1)
+/-/=	0/22/0	0/22/0	1/20/1	0/19/3	0/21/1	

2.2 MPolygon上各算法的实验结果

表 S14 MMODE-AID 与 5 个变种算法在 MPolygon 上获得的 IGDX 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MPolygon1	1.3281e-1 (1.23e-2) -	1.0984e-1 (5.40e-3) -	9.8342e-2 (2.09e-3) -	8.1255e-2 (1.21e-3) -	8.2998e-2 (7.33e-4) -	7.7373e-2 (9.87e-4)
MPolygon2	2.3903e-1 (3.60e-2) -	2.0672e-1 (1.14e-2) -	2.1288e-1 (9.16e-3) -	1.5472e-1 (3.57e-3) -	1.5777e-1 (7.21e-3) -	1.3698e-1 (1.92e-3)
MPolygon3	1.2392e-1 (7.68e-3) -	1.1067e-1 (3.15e-3) -	9.1041e-2 (1.26e-3) -	7.6240e-2 (6.84e-4) -	7.7173e-2 (7.79e-4) -	6.9354e-2 (7.87e-4)
MPolygon4	2.2261e-1 (1.77e-2) -	1.9232e-1 (5.88e-3) -	2.3286e-1 (1.30e-2) -	1.5661e-1 (3.22e-3) -	1.4945e-1 (2.91e-3) =	1.4987e-1 (3.24e-3)
MPolygon5	1.3261e-1 (9.29e-3) -	1.0958e-1 (3.00e-3) -	9.6944e-2 (1.14e-3) -	8.1460e-2 (5.82e-4) -	8.2534e-2 (6.53e-4) -	5.3772e-2 (1.76e-3)
MPolygon6	2.2412e-1 (2.20e-2) -	1.9124e-1 (6.30e-3) -	2.2797e-1 (1.08e-2) -	1.5773e-1 (2.69e-3) -	1.4898e-1 (4.37e-3) -	1.1795e-1 (1.44e-3)
MPolygon7	1.1406e-1 (6.28e-3) -	1.0038e-1 (2.83e-3) -	8.4655e-2 (6.82e-4) -	7.2087e-2 (5.24e-4) -	7.2879e-2 (4.51e-4) -	5.3042e-2 (1.18e-3)
MPolygon8	2.0198e-1 (2.27e-2) -	1.7274e-1 (4.12e-3) -	2.1390e-1 (7.69e-3) -	1.4165e-1 (2.21e-3) -	1.3358e-1 (2.81e-3) -	9.9456e-2 (1.11e-3)
MPolygon9	1.1647e-1 (7.76e-3) -	9.8407e-2 (2.58e-3) -	8.6905e-2 (7.37e-4) -	7.3627e-2 (4.68e-4) -	6.2174e-2 (4.99e-4) +	6.2695e-2 (1.41e-4)
MPolygon10	1.9935e-1 (1.56e-2) -	1.7432e-1 (4.44e-3) -	2.1830e-1 (1.29e-2) -	1.4448e-1 (2.17e-3) =	1.3782e-1 (2.19e-3) =	1.3626e-1 (1.36e-3)
MPolygon11	1.1929e-1 (5.68e-3) -	9.9386e-2 (2.56e-3) -	8.8130e-2 (1.69e-3) -	7.4298e-2 (5.50e-4) -	7.5115e-2 (4.57e-4) -	4.0653e-2 (4.53e-4)
MPolygon12	2.8231e-1 (1.83e-2) -	1.6483e-1 (6.82e-3) -	2.2910e-1 (1.73e-2) -	1.4585e-1 (1.85e-3) -	1.3834e-1 (3.61e-3) -	1.2450e-1 (1.39e-3)
+/-/=	0/12/0	0/12/0	0/12/0	0/11/1	1/9/2	

表 S15 MMODE-AID 与 5 个变种算法在 MPolygon 上获得的 IGDF 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MPolygon1	7.8238e-2 (5.48e-3) -	5.0158e-2 (1.61e-3) +	6.4828e-2 (1.98e-3) -	5.7083e-2 (6.40e-4) -	7.8132e-2 (5.17e-4) -	5.0454e-2 (1.79e-3)
MPolygon2	1.2947e-1 (2.39e-2) -	8.8249e-2 (3.58e-3) -	9.8324e-2 (3.36e-3) -	6.2915e-2 (1.02e-3) -	6.4632e-2 (1.86e-3) -	5.5454e-2 (1.15e-3)
MPolygon3	7.7493e-2 (3.72e-3) -	7.8231e-2 (2.41e-3) -	7.8294e-2 (1.85e-3) -	5.2683e-2 (4.06e-4) -	5.3188e-2 (5.87e-4) -	4.7902e-2 (5.30e-4)
MPolygon4	1.8504e-1 (1.38e-2) -	1.2841e-1 (2.83e-3) -	1.2854e-1 (5.53e-3) -	8.7222e-2 (1.04e-3) -	8.3212e-2 (8.89e-4) -	5.6773e-2 (7.08e-4)
MPolygon5	9.0283e-2 (4.74e-3) -	8.7382e-2 (2.93e-3) -	8.8420e-2 (1.74e-3) -	5.2478e-2 (4.29e-4) =	6.3999e-2 (5.78e-4) -	5.2238e-2 (1.15e-3)
MPolygon6	1.3238e-1 (1.82e-2) -	1.4632e-1 (2.64e-3) -	1.7329e-1 (4.83e-3) -	8.7396e-2 (9.26e-4) -	8.3697e-2 (1.24e-3) -	5.6404e-2 (7.56e-4)
MPolygon7	9.7392e-2 (3.84e-3) -	8.3829e-2 (2.94e-3) -	8.8294e-2 (1.36e-3) -	6.1726e-2 (3.72e-4) -	3.7083e-2 (3.65e-4) =	3.6964e-2 (7.32e-4)

表 S15 MMODE-AID 与 5 个变种算法在 MPolygon 上获得的 IGDF 均值和标准差 (续)

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MPolygon8	1.4762e-1 (1.30e-2) -	1.6392e-1 (2.42e-3) -	1.4820e-1 (3.28e-3) -	8.4812e-2 (1.08e-3) -	8.0594e-2 (1.04e-3) -	4.9786e-2 (6.82e-4)
MPolygon9	1.9320e-1 (3.14e-3) -	9.8302e-2 (2.83e-3) -	9.7392e-2 (1.84e-3) -	7.4119e-2 (4.77e-4) +	9.8506e-2 (3.90e-4) -	7.9685e-2 (8.38e-4)
MPolygon10	1.4725e-1 (1.32e-2) -	1.2744e-1 (2.35e-3) -	1.8291e-1 (6.34e-3) -	9.4420e-2 (1.20e-3) -	8.9851e-2 (1.07e-3) -	6.5977e-2 (6.42e-4)
MPolygon11	1.7422e-1 (3.46e-3) -	9.8257e-2 (2.77e-3) -	1.0703e-1 (1.06e-3) -	7.3758e-2 (4.67e-4) -	7.4322e-2 (4.59e-4) -	4.0421e-2 (4.73e-4)
MPolygon12	1.5732e-1 (1.74e-2) -	1.8426e-1 (3.73e-3) -	1.9242e-1 (5.24e-3) -	1.0212e-1 (1.17e-3) -	9.6811e-2 (1.09e-3) -	4.2403e-2 (6.25e-4)
+/-/=	0/12/0	1/11/0	0/12/0	1/10/1	0/11/1	

表 S16 MMODE-AID 与 5 个变种算法在 MPolygon 上获得的 PSP 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
MPolygon1	7.3152e+0 (6.46e-1) -	8.2885e+0 (4.93e-1) -	1.0042e+1 (3.34e-1) -	1.2303e+1 (1.83e-1) -	1.2017e+1 (1.12e-1) -	1.2924e+1 (1.94e-1)
MPolygon2	3.8256e+0 (5.93e-1) -	4.8338e+0 (2.67e-1) -	4.2013e+0 (2.03e-1) -	6.0568e+0 (1.50e-1) -	6.2510e+0 (2.95e-1) -	7.2942e+0 (1.16e-1)
MPolygon3	7.9742e+0 (5.91e-1) -	8.8244e+0 (3.52e-1) -	1.0284e+1 (1.28e-1) -	1.3111e+1 (1.18e-1) -	1.2945e+1 (1.34e-1) -	1.4413e+1 (1.56e-1)
MPolygon4	4.3403e+0 (3.57e-1) -	4.7293e+0 (1.48e-1) -	4.2925e+0 (2.93e-1) -	6.3838e+0 (1.32e-1) -	6.6905e+0 (1.34e-1) =	6.6721e+0 (1.25e-1)
MPolygon5	7.2198e+0 (6.27e-1) -	8.2128e+0 (2.96e-1) -	1.0173e+1 (1.81e-1) -	1.1575e+1 (8.74e-2) -	1.2111e+1 (9.81e-2) -	1.8593e+1 (1.08e-1)
MPolygon6	3.9374e+0 (4.27e-1) -	5.2164e+0 (2.91e-1) -	4.2418e+0 (2.25e-1) -	6.3207e+0 (1.08e-1) -	6.7122e+0 (1.95e-1) -	8.4778e+0 (1.24e-1)
MPolygon7	8.4339e+0 (4.81e-1) -	7.1632e+0 (2.29e-1) -	1.1047e+1 (8.22e-2) -	1.3862e+1 (1.01e-1) -	1.3721e+1 (8.48e-2) -	1.8850e+1 (1.47e-1)
MPolygon8	4.2871e+0 (5.27e-1) -	5.7204e+0 (3.48e-1) -	3.7180e+0 (1.19e-1) -	7.0562e+0 (1.11e-1) -	7.4855e+0 (1.57e-1) -	1.0040e+1 (1.39e-1)
MPolygon9	8.1648e+0 (7.14e-1) -	1.0124e+1 (2.73e-1) -	1.1417e+1 (7.13e-2) -	1.3571e+1 (8.63e-2) -	1.6082e+1 (9.05e-2) =	1.5948e+1 (5.15e-2)
MPolygon10	4.3041e+0 (3.73e-1) -	5.7149e+0 (1.26e-1) -	4.3031e+0 (2.83e-1) -	6.9201e+0 (1.05e-1) -	7.2530e+0 (1.15e-1) -	7.3389e+0 (1.19e-1)
MPolygon11	8.2031e+0 (4.92e-1) -	1.0013e+1 (2.74e-1) -	1.1246e+1 (3.18e-1) -	1.3450e+1 (9.99e-2) -	1.3312e+1 (8.00e-2) -	2.4580e+1 (4.84e-2)
MPolygon12	3.1973e+0 (4.42e-1) -	5.9254e+0 (1.48e-1) -	4.3325e+0 (3.52e-1) -	6.4574e+1 (8.70e-1) -	7.2131e+0 (1.87e-1) -	8.0318e+0 (2.79e-1)
+/-/=	0/12/0	0/12/0	0/12/0	0/12/0	0/10/2	

2.3 HYL上各算法的实验结果

表 S17 MMODE-AID 与 5 个变种算法在 HYL 上获得的 IGDX 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
HYL1	7.6420e-1 (1.27e-1) -	2.9202e-1 (2.73e-3) -	1.9045e-1 (7.16e-3) -	1.9848e-1 (2.08e-3) -	1.9901e-1 (2.17e-3) -	1.8634e-1 (9.61e-4)
HYL2	1.7011e+0 (2.85e-1) -	5.8308e-1 (9.03e-3) -	5.7488e-1 (8.22e-3) -	5.5749e-1 (4.34e-3) =	5.7923e-1 (4.35e-3) -	5.5735e-1 (4.35e-3)
HYL3	3.1713e+0 (4.26e-1) -	8.8837e-1 (1.31e-2) -	8.1257e-1 (9.82e-3) -	7.9489e-1 (7.24e-3) =	1.5174e+0 (2.20e+0) -	7.9303e-1 (8.18e-3)
HYL4	2.6056e+0 (1.47e-1) -	1.0558e+0 (1.22e-2) -	9.8792e-1 (8.99e-3) -	9.5005e-1 (6.63e-3) +	1.1769e+0 (1.05e+0) -	9.5609e-1 (6.21e-3)
HYL5	1.2935e+0 (2.67e-1) -	2.8849e-1 (5.13e-3) -	2.7935e-1 (2.11e-3) -	2.7177e-1 (2.63e-3) -	2.8210e-1 (2.12e-3) -	2.4612e-1 (2.04e-3)
+/-/=	0/5/0	0/5/0	0/5/0	1/2/2	0/5/0	

表 S18 MMODE-AID 与 5 个变种算法在 HYL 上获得的 IGDF 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
HYL1	3.6076e-1 (1.55e-1) -	1.7905e-2 (2.67e-3) -	1.8739e-2 (3.66e-3) -	1.9200e-2 (2.34e-3) -	1.6867e-2 (3.06e-3) =	1.6809e-2 (2.56e-3)
HYL2	1.3183e-1 (5.05e-2) -	1.4540e-2 (2.17e-3) =	1.6125e-2 (2.50e-3) -	1.7641e-2 (1.82e-3) -	1.4283e-2 (2.51e-3) =	1.4105e-2 (1.95e-3)
HYL3	1.1291e+0 (2.88e-1) -	1.5974e-1 (2.98e-3) -	1.3939e-1 (3.26e-3) -	1.5934e-1 (2.20e-3) -	1.4007e-1 (2.77e-3) -	1.3548e-1 (2.18e-3)
HYL4	7.8784e-1 (1.56e-1) -	1.6516e-1 (2.63e-3) -	1.4934e-1 (3.65e-3) -	1.4762e-1 (3.09e-3) -	1.4979e-1 (2.93e-3) -	1.4430e-1 (2.64e-3)
HYL5	6.4529e-1 (2.03e-1) -	4.2156e-2 (2.28e-3) -	3.0130e-2 (1.27e-3) =	2.9482e-2 (1.26e-3) =	3.0228e-2 (1.28e-3) =	2.9436e-2 (1.40e-3)
+/-/=	0/5/0	0/4/1	0/4/1	0/4/1	0/2/3	

表 S19 MMODE-AID 与 5 个变种算法在 HYL 上获得的 PSP 均值和标准差

测试问题	MMODE-AID-v1	MMODE-AID-v2	MMODE-AID-v3	MMODE-AID-v4	MMODE-AID-v5	MMODE-AID
HYL1	1.2671e+0 (4.19e-1) -	3.3692e+0 (2.35e-1) -	3.3518e+0 (5.21e-2) -	4.2305e+0 (8.83e-2) -	5.0171e+0 (7.89e-2) -	5.3665e+0 (2.77e-2)
HYL2	5.0036e-1 (1.44e-1) -	1.6362e+0 (1.26e-1) -	1.2435e+0 (2.71e-2) -	1.5901e+0 (1.53e-2) -	1.6842e+0 (1.46e-2) -	1.7942e+0 (1.37e-2)
HYL3	3.0231e-1 (1.23e-1) -	7.2608e-1 (1.35e-1) -	1.1481e+0 (1.38e-2) -	1.2551e+0 (2.09e-2) =	6.5620e-1 (3.69e-1) -	1.2608e+0 (1.34e-2)
HYL4	2.7457e-1 (4.33e-2) -	5.2481e-1 (1.36e-1) -	9.3240e-1 (9.29e-3) -	1.0450e+0 (6.19e-3) =	8.3655e-1 (1.95e-1) -	1.0456e+0 (6.80e-3)
HYL5	2.1636e+0 (4.10e-1) -	2.2563e+0 (2.25e-1) -	2.0603e+0 (3.87e-2) -	3.3457e+0 (3.73e-2) -	3.4624e+0 (3.31e-2) -	4.0628e+0 (3.62e-2)
+/-/=	0/5/0	0/5/0	0/5/0	0/3/2	0/5/0	

3 详细实验结果——策略有效性实验

3.1 CEC 2019 MMOPs 上各算法的实验结果

表 S20 4 个代表算法及对应变种算法在 CEC 2019 MMOPs 上获得的 IGDX 均值和标准差

测试问题	DN-NSGAI	DN-NSGAI-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO Ring P SO_SCD	MO Ring PSO_SCD-AID
MMF1	1.4011e-1 (2.11e-2) -	1.0906e-1 (1.33e-2)	4.5661e-2 (2.91e-3) -	3.6401e-2 (3.55e-3)	4.2639e-2 (2.60e-3) -	3.5893e-2 (2.28e-3)	6.5482e-2 (7.08e-3) -	5.1827e-2 (5.91e-3)
MMF2	1.0494e-1 (2.32e-2) -	5.7105e-2 (1.77e-2)	3.5249e-2 (1.44e-2) -	2.0326e-2 (4.57e-3)	4.5510e-2 (1.19e-2) -	2.5731e-2 (8.32e-3)	8.1122e-2 (3.46e-2) -	7.2944e-2 (4.37e-2)
MMF3	9.3320e-2 (2.44e-2) -	4.9000e-2 (1.43e-2)	3.1960e-2 (9.77e-3) -	1.7536e-2 (2.98e-3)	3.7185e-2 (7.55e-3) -	2.6946e-2 (7.12e-3)	7.7047e-2 (3.31e-2) -	5.6262e-2 (2.51e-2)
MMF4	9.6121e-2 (1.71e-2) -	2.8768e-2 (3.23e-3)	2.4919e-2 (1.49e-3) -	1.5139e-2 (1.64e-3)	2.0368e-2 (1.04e-3) =	2.0356e-2 (1.28e-3)	4.1537e-2 (5.40e-3) -	2.2497e-2 (2.43e-3)
MMF5	1.9960e-1 (2.12e-2) -	9.4673e-2 (7.30e-3)	7.5971e-2 (2.73e-3) -	6.8468e-2 (1.36e-3)	7.5566e-2 (5.37e-3) -	7.1364e-2 (4.25e-3)	1.0446e-1 (8.00e-3) -	8.5017e-2 (9.17e-3)
MMF6	1.6267e-1 (1.59e-2) -	7.7695e-2 (5.33e-3)	6.5061e-2 (2.64e-3) -	5.7725e-2 (1.18e-3)	6.2425e-2 (2.93e-3) +	6.8469e-2 (2.69e-3)	9.2411e-2 (1.07e-2) -	7.0156e-2 (7.67e-3)
MMF7	7.2417e-2 (1.01e-2) -	2.5855e-2 (3.69e-3)	2.4874e-2 (2.67e-3) -	1.3395e-2 (3.66e-3)	2.1113e-2 (2.60e-3) +	2.8125e-2 (2.93e-3)	3.6921e-2 (4.83e-3) -	2.3117e-2 (3.42e-3)
MMF8	2.2559e-1 (2.87e-2) -	1.6239e-1 (6.16e-2)	7.9480e-2 (1.62e-2) -	6.3363e-2 (1.26e-2)	9.0157e-2 (1.99e-2) -	8.5194e-2 (1.84e-2)	1.1034e-1 (2.72e-2) -	9.304e-2 (1.54e-2)
MMF9	2.4877e-2 (4.10e-3) -	5.9568e-3 (5.29e-4)	6.2638e-3 (2.52e-4) -	4.7781e-3 (3.31e-4)	4.5259e-3 (2.03e-4) -	4.1852e-3 (2.11e-4)	1.4260e-2 (6.51e-3) -	7.2221e-3 (1.00e-3)
MMF10	9.6387e-2 (2.90e-2) -	3.7475e-2 (1.09e-1)	1.3905e-2 (6.52e-2) -	5.5574e-3 (7.20e-2)	2.2154e-3 (4.08e-3) =	2.2398e-3 (8.94e-4)	2.1711e-2 (7.62e-3) -	1.0930e-2 (1.37e-3)
MMF11	2.5531e-2 (7.08e-3) -	5.2596e-3 (1.47e-3)	4.3861e-3 (2.36e-4) -	2.8019e-3 (1.79e-4)	3.3552e-3 (1.72e-4) -	3.0903e-3 (1.84e-4)	5.6668e-3 (3.12e-4) -	3.7899e-3 (1.73e-4)
MMF12	1.6870e-2 (5.10e-3) -	1.6108e-3 (2.59e-4)	1.7691e-3 (9.43e-5) -	1.3641e-3 (5.99e-5)	1.2899e-3 (9.78e-5) -	1.2168e-3 (7.74e-5)	5.1026e-3 (7.57e-4) -	1.4181e-3 (9.68e-5)
MMF13	9.0794e-2 (2.00e-2) -	2.3124e-2 (1.52e-3)	2.0216e-2 (2.88e-3) -	1.2672e-2 (1.11e-3)	1.8731e-2 (5.85e-4) -	1.3454e-2 (2.69e-3)	2.9648e-2 (1.16e-3) =	2.9465e-2 (1.92e-3)
MMF14	8.3230e-2 (5.40e-3) -	5.9529e-2 (4.63e-3)	4.2428e-2 (7.40e-4) -	3.2372e-2 (7.51e-4)	3.7914e-2 (2.73e-4) +	4.1619e-2 (2.57e-3)	6.1197e-2 (3.25e-3) -	4.5728e-2 (1.22e-3)

表 S20 4 个代表算法及对应变种算法在 CEC 2019 MMOPs 上获得的 IGDX 均值和标准差 (续)

测试问题	DN-NSGAI	DN-NSGAI-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_P SO_SCD	MO_Ring_PSO_SCD-AID
MMF15	8.3532e-2 (1.01e-2) -	3.6300e-2 (8.32e-4)	3.1297e-2 (5.89e-4) -	2.9552e-2 (5.61e-4)	2.8712e-2 (4.25e-4) -	2.5362e-2 (3.03e-3)	4.2530e-2 (1.49e-3) -	3.3415e-2 (8.57e-4)
MMF1_z	4.5119e-2 (8.08e-3) -	1.5119e-2 (8.02e-3)	3.3591e-2 (2.14e-3) -	2.9346e-2 (8.78e-3)	3.4645e-2 (4.26e-3) -	3.1295e-2 (3.37e-3)	4.8533e-2 (6.89e-3) -	3.9976e-2 (7.46e-3)
MMF1_e	1.2952e+0 (5.56e-1) -	4.2010e-1 (6.84e-2)	7.2500e-1 (1.63e-1) -	6.2090e-1 (1.48e-1)	1.0462e+0 (3.84e-1) -	9.3864e-1 (2.63e-1)	8.2344e-1 (3.47e-1) -	6.7100e-1 (2.35e-1)
MMF14_a	1.1549e-1 (9.90e-3) -	6.0739e-2 (1.89e-3)	5.7730e-2 (1.49e-3) -	4.8952e-2 (1.75e-3)	4.7918e-2 (5.74e-4) -	4.1936e-2 (4.68e-3)	6.8417e-2 (3.40e-3) -	5.4774e-2 (1.24e-3)
MMF15_a	1.0787e-1 (1.23e-2) -	4.1564e-2 (1.11e-3)	3.8093e-2 (7.31e-4) -	3.6689e-2 (6.96e-4)	3.2579e-2 (5.88e-4) -	3.0646e-2 (9.01e-3)	4.8070e-2 (2.05e-3) -	3.7197e-2 (7.47e-4)
Omni_test	1.5920e-1 (6.57e-2) -	8.5920e-2 (6.25e-2)	5.5336e-1 (1.59e-1) -	4.9695e-2 (3.90e-3)	1.1583e-1 (3.54e-2) -	8.8492e-2 (4.52e-3)	6.1923e-1 (1.55e-1) -	9.4819e-2 (8.24e-2)
SYM_RART simple	5.0916e-1 (6.58e-1) -	3.2617e-1 (6.48e-2)	5.8449e-2 (3.02e-3) -	4.7443e-2 (5.65e-3)	4.4559e-2 (4.03e-3) -	3.7106e-2 (3.62e-3)	2.7222e-1 (3.02e-1) -	2.3942e-1 (3.98e-1)
SYM_RART rotated	9.7049e-1 (8.15e-1) -	3.9752e-1 (5.07e-2)	2.6119e-1 (4.45e-1) -	9.5171e-2 (2.17e-2)	4.5484e-1 (5.71e-1) -	3.5457e-2 (4.42e-3)	2.5881e-1 (2.45e-1) +	3.9806e-1 (9.00e-1)
+/-/=	0/22/0		0/22/0		3/17/2		1/20/1	

表 S21 4 个代表算法及对应变种算法在 CEC 2019 MMOPs 上获得的 IGDF 均值和标准差

测试问题	DN-NSGAI	DN-NSGAI-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_P SO_SCD	MO_Ring_PSO_SCD-AID
MMF1	3.9670e-2 (1.97e-2) -	7.8978e-3 (1.26e-3)	3.8082e-3 (2.75e-4) -	2.4909e-3 (1.91e-4)	3.2159e-3 (1.55e-4) -	2.5694e-3 (9.53e-5)	4.4866e-3 (3.98e-4) -	2.6389e-3 (8.39e-5)
MMF2	1.6876e-1 (5.05e-2) -	2.1559e-2 (9.23e-3)	1.3565e-2 (7.80e-3) -	8.2134e-3 (2.25e-3)	1.8246e-2 (5.55e-3) -	8.4382e-3 (7.94e-4)	4.3886e-2 (1.47e-2) -	1.3237e-2 (2.14e-3)
MMF3	1.4746e-1 (5.10e-2) -	1.6687e-2 (7.25e-3)	1.1158e-2 (2.68e-3) -	7.0173e-3 (1.49e-3)	1.3184e-2 (3.03e-3) -	9.4683e-3 (7.64e-4)	3.7929e-2 (1.39e-2) -	1.1051e-2 (1.66e-3)
MMF4	2.1680e-2 (7.21e-3) -	3.1203e-3 (1.18e-4)	3.7432e-3 (2.58e-4) -	2.3180e-3 (1.63e-4)	3.2896e-3 (1.43e-4) -	2.1678e-3 (8.15e-5)	3.7503e-3 (2.31e-4) -	2.5419e-3 (6.57e-5)
MMF5	4.0477e-2 (1.92e-2) -	3.1803e-3 (1.50e-4)	3.5755e-3 (2.55e-4) -	2.4190e-3 (1.75e-4)	3.2393e-3 (1.58e-4) -	2.4516e-3 (8.46e-5)	3.9793e-3 (2.67e-4) -	2.6247e-3 (8.41e-5)
MMF6	2.5948e-2 (1.09e-2) -	3.1239e-3 (1.23e-4)	3.4938e-3 (2.44e-4) -	2.5082e-3 (1.90e-4)	3.1986e-3 (1.25e-4) -	2.4158e-3 (7.38e-5)	3.8315e-3 (2.69e-4) -	2.5315e-3 (5.80e-5)
MMF7	1.4596e-2 (2.34e-3) -	3.2806e-3 (1.61e-4)	3.9173e-3 (2.65e-4) -	2.2920e-3 (1.21e-4)	3.1893e-3 (1.46e-4) -	2.4873e-3 (7.37e-5)	3.7897e-3 (3.66e-4) -	2.5438e-3 (5.31e-5)
MMF8	1.8065e-2 (4.24e-3) -	4.4568e-3 (3.90e-4)	3.6830e-3 (1.97e-4) -	2.7141e-3 (2.55e-4)	3.7445e-3 (2.08e-4) -	3.1602e-3 (3.42e-4)	4.7112e-3 (2.10e-4) -	3.1101e-3 (1.23e-4)
MMF9	4.4290e-2 (7.10e-3) -	1.3551e-2 (1.40e-3)	1.5641e-2 (1.20e-3) -	9.8898e-3 (2.50e-3)	1.3753e-2 (1.02e-3) -	1.0274e-2 (7.94e-4)	1.6996e-2 (1.60e-3) -	9.8800e-3 (4.64e-4)
MMF10	2.8837e-1 (3.71e-2) -	3.8962e-2 (8.86e-2)	2.2994e-2 (5.24e-2) -	1.0011e-2 (2.38e-2)	9.9863e-3 (7.50e-3) -	7.5374e-3 (6.74e-4)	6.3879e-2 (1.18e-2) -	7.3608e-3 (5.46e-4)
MMF11	7.1899e-2 (1.39e-2) -	2.0616e-2 (2.68e-3)	1.5620e-2 (1.97e-3) -	8.6798e-3 (5.64e-3)	1.3529e-2 (1.06e-3) -	8.4613e-3 (4.96e-4)	1.6571e-2 (1.58e-3) -	1.0008e-2 (5.22e-4)
MMF12	4.7366e-2 (1.42e-2) -	5.0698e-3 (1.39e-3)	2.8630e-3 (2.02e-4) -	1.6746e-3 (1.44e-4)	2.4029e-3 (1.04e-4) -	1.6357e-3 (1.36e-4)	1.2587e-2 (2.26e-3) -	2.1831e-3 (8.65e-5)
MMF13	8.8169e-2 (1.95e-2) -	1.3707e-2 (7.86e-4)	1.6215e-2 (1.57e-3) -	8.2871e-3 (1.04e-3)	1.5104e-2 (2.04e-3) -	8.6608e-3 (7.64e-4)	1.3172e-2 (6.12e-4) -	8.8212e-3 (1.83e-4)
MMF14	1.3300e-1 (9.95e-3) -	8.8911e-2 (5.44e-3)	6.9447e-2 (1.16e-3) -	4.6534e-2 (8.44e-4)	6.4308e-2 (1.00e-3) -	5.1647e-2 (1.73e-3)	8.1788e-2 (2.47e-3) -	5.7919e-2 (8.26e-4)
MMF15	1.8803e-1 (2.08e-2) -	7.6065e-2 (2.07e-3)	6.7836e-2 (1.63e-3) -	5.7537e-2 (2.71e-3)	5.0829e-2 (7.20e-4) +	5.4162e-2 (2.47e-3) -	8.3776e-2 (2.33e-3) -	5.8534e-2 (1.07e-3)
MMF1_z	2.9217e-2 (1.04e-2) -	3.1678e-3 (1.64e-4)	3.7599e-3 (2.31e-4) -	2.2046e-3 (1.58e-4)	3.1264e-3 (1.71e-4) -	1.4682e-3 (8.46e-5)	4.2672e-3 (3.10e-4) -	2.5911e-3 (1.02e-4)
MMF1_e	1.3683e-1 (7.23e-2) -	5.7962e-2 (2.66e-3)	3.0328e-2 (1.69e-2) -	7.8103e-3 (4.50e-4)	5.3384e-3 (1.46e-3) -	4.7458e-3 (6.37e-4)	1.1373e-2 (1.49e-3) =	1.1201e-2 (1.58e-3)
MMF14_a	1.5964e-1 (1.40e-2) -	7.5983e-2 (1.84e-3)	7.4047e-2 (1.77e-3) -	6.6925e-2 (1.06e-3)	6.5888e-2 (1.04e-3) -	6.2174e-2 (1.13e-3)	8.0618e-2 (2.37e-3) -	5.9163e-2 (1.02e-3)
MMF15_a	2.1147e-1 (2.03e-2) -	7.6520e-2 (1.77e-3)	7.1379e-2 (1.46e-3) -	6.7670e-2 (1.51e-3)	6.3126e-2 (9.20e-4) -	5.6941e-2 (1.06e-3)	8.4908e-2 (2.61e-3) -	6.1339e-2 (8.30e-4)
Omni_test	1.0203e-1 (1.71e-2) -	1.3123e-2 (3.47e-3)	1.2989e-2 (2.72e-3) -	1.0506e-2 (4.84e-3)	1.1181e-2 (2.56e-3) =	1.1164e-2 (2.58e-3)	4.2336e-2 (6.09e-3) -	1.0550e-2 (1.71e-3)
SYM_RART simple	6.2811e-2 (9.74e-3) -	2.3481e-2 (3.18e-3)	1.5941e-2 (1.94e-3) -	1.0799e-2 (2.89e-3)	1.4483e-2 (1.29e-3) -	1.2468e-2 (1.75e-3)	3.6033e-2 (4.33e-3) -	1.4426e-2 (2.25e-3)
SYM_RART rotated	9.8232e-2 (1.66e-2) -	1.7655e-2 (2.12e-3)	1.8591e-2 (1.69e-3) =	1.8227e-2 (3.65e-3)	1.6273e-2 (1.86e-3) -	1.3605e-2 (1.72e-3)	4.4384e-2 (6.38e-3) -	2.3654e-2 (3.25e-3)
+/-/=	0/22/0		0/21/1		1/20/1		0/21/1	

表 S22 4 个代表算法及对应变种算法在 CEC 2019 MMOPs 上获得的 PSP 均值和标准差

测试问题	DN-NSGAI	DN-NSGAI-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_P SO_SCD	MO_Ring_PSO SCD-AID
MMF1	6.8404e+0 (1.23e+0) -	9.1162e+0 (1.22e+0)	1.8920e+1 (1.44e+0) -	2.5319e+1 (8.98e-1)	2.3367e+1 (1.37e+0) -	2.7310e+1 (8.34e-1)	1.5242e+1 (1.76e+0) -	1.9236e+1 (1.51e+0)
MMF2	9.3038e+0 (2.13e+0) -	1.6310e+1 (2.60e+0)	2.8355e+1 (8.67e+0) -	4.8585e+1 (1.16e+0)	2.1504e+1 (6.42e+0) -	3.8506e+1 (5.49e+0)	1.2298e+1 (4.13e+0) -	1.3633e+1 (4.25e+0)
MMF3	1.0549e+1 (3.16e+0) -	1.9261e+1 (1.90e+0)	3.0456e+1 (9.07e+0) -	5.6254e+1 (1.04e+0)	2.6780e+1 (5.59e+0) -	3.6035e+1 (6.12e+0)	1.2594e+1 (6.03e+0) -	1.7116e+1 (5.63e+0)
MMF4	1.0336e+1 (1.79e+0) -	3.4673e+1 (1.29e+0)	4.0118e+1 (2.34e+0) -	6.3167e+1 (2.27e+0)	4.9046e+1 (2.54e+0) =	4.9119e+1 (1.05e+0)	2.4066e+1 (3.07e+0) -	4.4435e+1 (1.32e-1)
MMF5	4.6791e+0 (7.06e-1) -	1.0332e+1 (1.37e-1)	1.3135e+1 (4.78e-1) -	1.4527e+1 (4.30e-1)	1.3231e+1 (9.45e-1) -	1.3673e+1 (5.74e-1)	9.5291e+0 (7.70e-1) -	1.0523e+1 (6.16e-1)
MMF6	5.6578e+0 (6.76e-1) -	1.2317e+1 (1.33e-1)	1.5338e+1 (6.26e-1) -	1.7280e+1 (4.56e-1)	1.6005e+1 (7.56e-1) +	1.4604e+1 (5.75e-1)	1.0816e+1 (1.19e+0) -	1.4238e+1 (1.03e+0)
MMF7	1.3534e+1 (2.05e+0) -	3.8490e+1 (1.62e+0)	4.0105e+1 (4.05e+0) -	7.2273e+1 (2.36e+0)	4.6781e+1 (5.29e+0) +	3.4701e+1 (2.98e+0) -	2.7056e+1 (3.89e+0) -	4.3115e+1 (1.11e+0)
MMF8	4.4259e+0 (5.50e-1) -	6.1361e+0 (3.73e-1)	1.2409e+1 (2.46e+0) -	1.5737e+1 (6.64e-1)	1.1043e+1 (2.76e+0) -	1.1706e+1 (1.53e+0)	9.0563e+0 (2.13e+0) -	1.0697e+1 (1.41e+0)
MMF9	4.0188e+1 (6.21e+0) -	1.6071e+2 (1.70e+0)	1.5676e+2 (6.08e+0) -	2.0912e+2 (7.05e+0)	2.1128e+2 (9.58e+0) -	2.3583e+2 (7.14e+0)	7.0109e+1 (1.80e+1) -	1.3684e+2 (1.03e+1)
MMF10	1.0258e+1 (3.50e+0) -	2.6187e+1 (2.01e+0)	6.9578e+1 (9.53e+1) -	1.7415e+2 (3.23e+1)	4.2418e+2 (1.64e+2) =	4.3453e+2 (1.84e+1)	4.6042e+1 (1.68e+1) -	9.0589e+1 (1.12e+1)
MMF11	3.8399e+1 (9.83e+0) -	1.8939e+2 (1.23e+0)	2.2339e+2 (1.21e+1) -	3.5572e+2 (1.08e+1)	2.9803e+2 (1.58e+1) -	3.2115e+2 (1.42e+1)	1.7606e+2 (1.01e+1) -	2.6272e+2 (1.03e+1)
MMF12	5.8557e+1 (1.61e+1) -	6.1830e+2 (4.78e+1)	5.5686e+2 (3.10e+1) -	7.3247e+2 (2.10e+1)	6.7965e+2 (6.18e+1) -	8.1967e+2 (4.63e+1)	1.9592e+2 (3.07e+1) -	7.0494e+2 (2.96e+1)
MMF13	1.1011e+1 (2.48e+0) -	4.2630e+1 (2.94e+0)	4.7414e+1 (1.21e+0) -	7.8627e+1 (1.35e-1)	5.1370e+1 (1.70e+0) -	7.4260e+1 (7.45e-1)	3.3502e+1 (1.34e+0) -	3.3521e+1 (4.01e-1)
MMF14	1.2012e+1 (7.59e-1) -	1.5869e+1 (5.73e-1)	2.3176e+1 (4.11e-1) -	3.0851e+1 (3.15e-1)	2.6335e+1 (1.90e-1) +	2.1019e+1 (2.56e-1)	1.6283e+1 (8.64e-1) -	2.1229e+1 (4.40e-1)
MMF15	1.1928e+1 (1.44e+0) -	2.6883e+1 (7.63e-1)	3.1463e+1 (6.01e-1) -	3.3822e+1 (5.67e-1)	3.3435e+1 (5.20e-1) -	3.9324e+1 (4.28e-1)	2.3437e+1 (8.17e-1) -	2.9103e+1 (3.25e-1)
MMF1_z	2.1178e+1 (7.72e-1) -	6.5265e+1 (4.09e-1)	2.9333e+1 (4.48e-1) -	3.4002e+1 (2.64e-1)	2.8171e+1 (3.52e+0) -	3.1638e+1 (3.47e-1)	1.8617e+1 (7.23e-1) -	2.4233e+1 (5.28e-1)
MMF1_e	6.3596e-1 (1.04e+0) -	2.2897e+0 (6.31e-1)	1.3261e+0 (5.02e-1) -	1.6021e+0 (3.91e-1)	9.3823e-1 (4.27e-1) -	1.0629e+0 (4.12e-1)	1.0828e+0 (8.53e-1) -	1.4850e+0 (5.15e-1)
MMF14_a	7.9373e+0 (1.74e+0) -	1.6321e+1 (1.14e+0)	1.6714e+1 (1.90e+0) =	1.6880e+1 (1.67e+0)	2.0862e+1 (2.51e-1) -	2.3834e+1 (1.09e+0)	1.4563e+1 (2.74e+0) -	1.8038e+1 (2.64e+0)
MMF15_a	9.0560e+0 (5.49e-1) -	2.3453e+1 (2.85e-1)	2.5299e+1 (3.92e-1) -	2.7197e+1 (3.29e-1)	2.9704e+1 (5.50e-1) -	3.1467e+1 (3.52e-1)	2.0102e+1 (5.47e-1) -	2.6315e+1 (2.28e-1)
Omni_test	6.2012e+0 (3.89e-1) -	1.1446e+1 (3.03e-1)	1.8047e+0 (5.82e-1) -	2.0112e+1 (5.67e-1)	8.1506e+0 (2.12e+0) -	1.1201e+1 (1.24e+0)	1.5024e+0 (5.63e-1) -	1.0274e+1 (2.22e-1)
SYM_RART simple	1.9036e+0 (7.49e-1) -	2.9594e+0 (6.54e-1)	1.6930e+1 (8.71e-1) =	1.7015e+1 (9.57e-1)	2.2308e+1 (2.07e+0) -	2.6873e+1 (5.93e-1)	3.5539e+0 (1.71e+0) -	4.1649e+0 (6.42e-1)
SYM_RART rotated	1.0119e+0 (3.29e-1) -	2.5032e+0 (2.01e-1)	3.8084e+0 (3.56e+0) -	1.0450e+1 (1.07e+0)	2.1688e+0 (4.15e+0) -	2.7820e+1 (3.54e-1)	3.8500e+0 (1.27e+0) +	2.5080e+0 (1.70e-1)
+/-/=	0/22/0		0/20/2		3/17/2		1/21/0	

3.2 MPolygon上各算法的实验结果

表 S23 4 个代表算法及对应变种算法在 MPolygon 上获得的 IGDX 均值和标准差

测试问题	DN-NSGAI	DN-NSGAI-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_P SO_SCD	MO_Ring_PSO SCD-AID
MPolygon1	1.3281e-1 (1.23e-2) -	9.4312e-2 (2.43e-3)	9.6829e-2 (2.75e-3) -	8.5152e-2 (1.66e-3)	8.0304e-2 (1.21e-3) =	8.0298e-2 (1.71e-3)	1.0984e-1 (5.40e-3) -	7.9319e-2 (1.39e-3)
MPolygon2	2.3903e-1 (3.60e-2) -	1.4789e-1 (6.71e-3)	1.7734e-1 (4.92e-3) =	1.7013e-1 (1.06e-2)	1.5561e-1 (3.48e-3) -	5.2469e-2 (1.67e+0)	2.0672e-1 (1.14e-2) -	1.5707e-1 (3.59e-3)
MPolygon3	1.2392e-1 (7.68e-3) -	9.9708e-2 (2.40e-3)	9.1400e-2 (2.18e-3) -	8.1277e-2 (1.08e-3)	7.5983e-2 (7.27e-4) +	7.9518e-2 (9.27e-4)	1.1067e-1 (3.15e-3) -	7.5491e-2 (7.11e-4)
MPolygon4	2.2261e-1 (1.77e-2) -	1.5634e-1 (4.11e-3)	1.8208e-1 (4.47e-3) -	1.2287e-1 (3.75e-3)	1.5955e-1 (2.87e-3) -	9.5874e-2 (1.25e-3)	1.9232e-1 (5.88e-3) -	1.2230e-1 (3.60e-3)
MPolygon5	1.3261e-1 (9.29e-3) -	1.0037e-1 (1.84e-3)	9.7452e-2 (1.87e-3) -	8.7151e-2 (8.40e-4)	8.1634e-2 (7.02e-4) -	7.8280e-2 (8.30e-4)	1.0958e-1 (3.00e-3) -	8.1204e-2 (7.48e-4)
MPolygon6	2.2412e-1 (2.20e-2) -	1.5709e-1 (3.75e-3)	1.8021e-1 (4.91e-3) -	1.2294e-1 (4.11e-3)	1.6039e-1 (2.55e-3) -	1.1670e-1 (2.95e-3)	1.9124e-1 (6.30e-3) -	1.3123e-1 (3.08e-3)
MPolygon7	1.1406e-1 (6.28e-3) -	9.1971e-2 (1.89e-3)	8.5391e-2 (1.29e-3) -	7.2896e-2 (7.60e-4)	7.2021e-2 (5.71e-4) -	6.5247e-2 (6.83e-4)	1.0038e-1 (2.83e-3) -	7.1721e-2 (5.97e-4)

表 S23 4 个代表算法及对应变种算法在 MPolygon 上获得的 IGDX 均值和标准差 (续)

测试问题	DN-NSGAII	DN-NSGAII-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_P SO_SCD	MO_Ring_PSO_SCD-AID
MPolygon8	2.0198e-1 (2.27e-2) -	1.4092e-1 (2.64e-3)	1.6271e-1 (4.06e-3) -	1.0100e-1 (5.24e-3)	1.4265e-1 (2.32e-3) -	1.1469e-1 (2.21e-3)	1.7274e-1 (4.12e-3) -	1.3559e-1 (2.64e-3)
MPolygon9	1.1647e-1 (7.76e-3) -	9.2300e-2 (1.54e-3)	8.7499e-2 (1.31e-3) -	6.8480e-2 (7.55e-4)	7.3734e-2 (4.99e-4) -	7.1607e-2 (2.72e-4)	9.8407e-2 (2.58e-3) -	7.3516e-2 (6.90e-4)
MPolygon10	1.9935e-1 (1.56e-2) -	1.2567e-1 (3.42e-3)	1.6490e-1 (2.73e-3) -	1.0512e-1 (2.13e-3)	1.4598e-1 (3.09e-3) -	9.1120e-2 (2.54e-3)	1.7432e-1 (4.44e-3) -	1.1832e-1 (1.86e-3)
MPolygon11	1.1929e-1 (5.68e-3) -	9.3896e-2 (1.72e-3)	8.8478e-2 (1.76e-3) =	8.8218e-2 (6.48e-4)	7.4741e-2 (5.74e-4) -	7.1759e-2 (4.67e-4)	9.9386e-2 (2.56e-3) -	7.4223e-2 (4.93e-4)
MPolygon12	2.0478e-1 (1.77e-2) -	1.4568e-1 (4.36e-3)	1.6757e-1 (4.17e-3) -	1.0628e-1 (4.68e-3)	1.4913e-1 (2.24e-3) -	1.2505e-1 (2.33e-3)	1.7447e-1 (5.15e-3) -	1.1015e-1 (2.13e-3)
+/-/=	0/12/0		0/10/2		1/10/1		0/12/0	

表 S24 4 个代表算法及对应变种算法在 MPolygon 上获得的 IGDF 均值和标准差

测试问题	DN-NSGAII	DN-NSGAII-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_P SO_SCD	MO_Ring_PSO_SCD-AID
MPolygon1	7.0907e-2 (5.13e-3) -	6.2694e-2 (3.72e-3)	6.1180e-2 (1.52e-3) -	5.4333e-2 (1.38e-3)	5.4820e-2 (1.29e-3) -	4.9928e-2 (8.62e-4)	4.9958e-2 (1.01e-3) -	4.1756e-2 (5.91e-4)
MPolygon2	1.0702e-1 (2.35e-2) -	7.5097e-2 (1.72e-2)	7.7658e-2 (2.51e-3) -	6.1296e-2 (1.61e-3)	7.0860e-2 (1.45e-3) -	6.1987e-2 (7.56e-4)	8.5684e-2 (3.85e-3) -	7.6278e-2 (1.83e-3)
MPolygon3	7.8492e-2 (3.61e-3) -	5.9400e-2 (1.69e-3)	6.8120e-2 (1.34e-3) -	5.1830e-2 (9.42e-4)	6.1304e-2 (7.77e-4) -	5.7631e-2 (6.74e-4)	7.5893e-2 (2.40e-3) -	5.7201e-2 (6.35e-4)
MPolygon4	1.3743e-1 (1.62e-2) -	8.3854e-2 (3.41e-3)	1.0716e-1 (2.23e-3) -	9.9186e-2 (1.87e-3)	9.8765e-2 (2.11e-3) -	8.5686e-2 (1.27e-3)	1.1068e-1 (2.80e-3) =	1.0882e-1 (1.74e-3)
MPolygon5	9.5490e-2 (4.98e-3) -	6.6859e-2 (3.16e-2)	8.2220e-2 (1.36e-3) -	7.4722e-2 (1.05e-3)	7.4236e-2 (1.13e-3) -	6.7340e-2 (7.21e-4)	8.4795e-2 (2.38e-3) -	6.8274e-2 (5.89e-4)
MPolygon6	1.3964e-1 (1.74e-2) -	7.3116e-2 (3.05e-3)	1.0669e-1 (2.28e-3) -	9.8505e-2 (1.65e-3)	9.8479e-2 (1.26e-3) -	8.0579e-2 (1.51e-3)	1.1117e-1 (2.96e-3) -	1.0268e-1 (1.53e-3)
MPolygon7	9.1676e-2 (3.60e-3) -	5.5851e-2 (5.48e-3)	7.9119e-2 (1.32e-3) -	7.2362e-2 (8.42e-4)	7.1831e-2 (8.61e-4) -	5.3486e-2 (5.38e-4)	8.5576e-2 (2.12e-3) -	6.5748e-2 (6.06e-4)
MPolygon8	1.3292e-1 (1.50e-2) -	7.4053e-2 (5.12e-3)	1.0411e-1 (2.22e-3) -	9.6572e-2 (1.43e-3)	9.5803e-2 (1.57e-3) -	7.0898e-2 (1.33e-3)	1.0799e-1 (2.58e-3) -	9.9848e-2 (1.14e-3)
MPolygon9	1.0024e-1 (3.89e-3) -	9.9733e-2 (7.31e-3)	8.7819e-2 (1.32e-3) -	8.0386e-2 (1.05e-3)	7.9696e-2 (1.01e-3) =	7.9701e-2 (5.20e-4)	9.0374e-2 (2.11e-3) -	7.2424e-2 (5.87e-4)
MPolygon10	1.4410e-1 (1.17e-2) -	9.9308e-2 (2.37e-2)	1.1473e-1 (1.92e-3) -	1.0700e-1 (1.61e-3)	1.0634e-1 (1.47e-3) -	8.2039e-2 (1.38e-3)	1.1884e-1 (2.44e-3) -	1.1061e-1 (1.42e-3)
MPolygon11	1.1023e-1 (3.85e-3) -	9.8397e-2 (1.75e-3)	9.4891e-2 (1.60e-3) -	8.6350e-2 (1.09e-3)	8.6052e-2 (1.08e-3) -	7.5832e-2 (5.36e-4)	9.8887e-2 (2.11e-3) -	7.8262e-2 (6.36e-4)
MPolygon12	1.5898e-1 (1.87e-2) -	1.1768e-1 (6.10e-3)	1.2476e-1 (2.35e-3) -	1.1654e-1 (1.68e-3)	1.1649e-1 (2.00e-3) -	8.5647e-2 (1.84e-3)	1.2841e-1 (3.09e-3) =	1.1999e-1 (1.87e-3)
+/-/=	0/12/0		0/12/0		0/11/1		0/10/2	

表 S25 4 个代表算法及对应变种算法在 MPolygon 上获得的 PSP 均值和标准差

测试问题	DN-NSGAII	DN-NSGAII-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_P SO_SCD	MO_Ring_PSO_SCD-AID
MPolygon1	7.4482e+0 (6.50e-1) -	1.0187e+1 (5.78e-1)	1.0229e+1 (2.92e-1) -	1.1172e+1 (2.14e-1)	1.2451e+1 (1.83e-1) =	1.2448e+1 (2.37e-1)	9.0157e+0 (4.32e-1) -	1.2527e+1 (2.24e-1)
MPolygon2	3.8663e+0 (5.89e-1) -	6.5276e+0 (2.14e-1)	5.6307e+0 (1.57e-1) -	5.8019e+0 (2.07e-1)	6.4101e+0 (1.51e-1) -	1.5360e+1 (1.41e-1)	4.8338e+0 (2.67e-1) -	6.3490e+0 (2.13e-1)
MPolygon3	7.9353e+0 (5.01e-1) -	1.0012e+1 (7.10e-1)	1.0933e+1 (2.53e-1) =	1.0986e+1 (1.49e-1)	1.3157e+1 (1.28e-1) +	1.2562e+1 (1.48e-1)	9.0357e+0 (2.56e-1) -	1.3213e+1 (1.62e-1)
MPolygon4	4.2990e+0 (3.48e-1) -	6.3589e+0 (2.50e-1)	5.4836e+0 (1.35e-1) -	8.1075e+0 (1.42e-1)	6.2668e+0 (1.11e-1) -	1.0371e+1 (2.74e-1)	5.1991e+0 (1.63e-1) -	7.4906e+0 (1.33e-1)
MPolygon5	7.4918e+0 (5.23e-1) -	9.8904e+0 (5.09e-1)	1.0253e+1 (1.95e-1) -	1.1417e+1 (1.22e-1)	1.2246e+1 (1.05e-1) -	1.2671e+1 (1.34e-1)	9.1256e+0 (2.53e-1) -	1.2274e+1 (1.09e-1)
MPolygon6	4.2754e+0 (4.67e-1) -	6.2645e+0 (4.01e-1)	5.5311e+0 (1.50e-1) -	8.1268e+0 (1.24e-1)	6.2344e+0 (9.88e-2) -	8.5652e+0 (6.62e-2)	5.2284e+0 (1.69e-1) -	7.5878e+0 (1.02e-1)
MPolygon7	8.7492e+0 (4.81e-1) -	9.9405e+0 (2.98e-1)	1.1703e+1 (1.77e-1) =	1.1711e+1 (9.50e-1) =	1.3880e+1 (1.11e-1) =	1.3896e+1 (1.34e-1)	9.9582e+0 (2.82e-1) -	1.3906e+1 (1.29e-1)
MPolygon8	4.8853e+0 (5.25e-1) -	6.9604e+0 (4.54e-1)	6.0590e+0 (1.53e-1) -	9.6810e+0 (1.68e-1)	7.0011e+0 (1.12e-1) -	8.7178e+0 (2.47e-1)	5.7816e+0 (1.39e-1) -	7.3684e+0 (1.30e-1)
MPolygon9	8.5768e+0 (5.02e-1) -	9.9443e+0 (3.25e-1)	1.1231e+1 (1.71e-1) -	1.4508e+1 (9.78e-2)	1.3561e+1 (9.13e-2) -	1.3946e+1 (8.25e-2)	1.0157e+1 (2.68e-1) =	1.0241e+1 (1.20e-1)

表 S25 4 个代表算法及对应变种算法在 MPolygon 上获得的 PSP 均值和标准差 (续)

测试问题	DN-NSGAII	DN-NSGAII-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_PSO_SCD	MO_Ring_PSO_SCD-AID
MPolygon10	4.9496e+0 (3.99e-1) -	7.7363e+0 (3.20e-1)	5.9657e+0 (1.01e-1) -	9.5070e+0 (2.84e-1)	6.8528e+0 (1.48e-1) -	1.0939e+1 (2.88e-1)	5.7363e+0 (1.42e-1) -	8.4482e+0 (1.23e-1)
MPolygon11	8.3276e+0 (4.07e-1) -	1.0623e+1 (3.92e-1)	1.1107e+1 (2.25e-1) -	1.1321e+1 (2.08e-1)	1.3370e+1 (1.03e-1) -	1.3935e+1 (1.06e-1)	1.0057e+1 (2.54e-1) -	1.3428e+1 (1.01e-1)
MPolygon12	4.7269e+0 (4.12e-1) -	6.8446e+0 (3.26e-1)	5.8712e+0 (1.50e-1) -	9.4077e+0 (1.29e-1)	6.7055e+0 (1.01e-1) -	7.9844e+0 (2.75e+0)	5.7311e+0 (1.68e-1) -	9.0499e+0 (1.11e-1)
+/-/=	0/12/0		0/10/2		1/9/2		0/11/1	

3.3 HYL上各算法的实验结果

表 S26 4 个代表算法及对应变种算法在 HYL 上获得的 IGDX 均值与标准差

测试问题	DN-NSGAII	DN-NSGAII-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_PSO_SCD	MO_Ring_PSO_SCD-AID
HYL1	3.7149e-1 (4.23e-2) -	2.7526e-1 (6.03e-1)	2.1027e-1 (6.41e-3) -	1.8320e-1 (2.68e-3)	1.8771e-1 (1.64e-3) -	1.7534e-1 (2.65e-3)	2.4477e-1 (8.69e-3) -	2.1305e-1 (5.12e-3)
HYL2	7.0358e-1 (2.75e-2) -	6.9358e-1 (7.10e-2)	6.3955e-1 (1.31e-2) -	5.8513e-1 (8.23e-3)	5.6605e-1 (4.43e-3) -	5.0184e-1 (1.39e-2)	9.5182e-1 (1.53e-1) -	6.3207e-1 (1.25e-2)
HYL3	9.2033e-1 (1.93e-2) -	8.0176e-1 (1.31e-2)	8.5886e-1 (1.39e-2) -	6.1513e-1 (8.23e-3)	8.4497e-1 (1.57e-2) -	7.5235e-1 (2.69e-2)	1.0685e+0 (1.38e-1) -	8.3207e-1 (1.25e-2)
HYL4	1.9207e+0 (1.88e-1) -	9.1374e-1 (5.09e-2)	1.0411e+0 (1.53e-2) -	9.7403e-1 (1.38e-2)	9.9901e-1 (1.07e-2) -	9.2864e-1 (2.53e-2)	1.8312e+0 (4.56e-1) -	9.0740e-1 (2.14e-2)
HYL5	3.8397e-1 (3.87e-2) -	2.6725e-1 (1.50e-2)	2.8020e-1 (5.99e-3) -	1.2776e-1 (1.98e-3)	2.4805e-1 (2.77e-3) -	1.7269e-1 (1.29e-3)	4.3024e-1 (4.61e-2) -	2.1043e-1 (2.03e-2)
+/-/=	0/5/0		0/5/0		0/5/0		0/5/0	

表 S27 4 个代表算法及对应变种算法在 HYL 上获得的 IGDF 均值与标准差

测试问题	DN-NSGAII	DN-NSGAII-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_PSO_SCD	MO_Ring_PSO_SCD-AID
HYL1	4.2304e-2 (1.12e-2) -	3.3801e-2 (8.76e-5)	3.1238e-2 (6.60e-3) -	2.9422e-2 (6.66e-3)	3.8994e-2 (9.25e-3) =	3.8985e-2 (3.28e-3)	2.8638e-2 (7.15e-3) -	1.7650e-2 (2.75e-3)
HYL2	4.2465e-2 (7.34e-3) -	1.3529e-2 (3.15e-5)	3.0553e-2 (5.76e-3) -	2.0969e-2 (5.04e-3)	4.0851e-2 (9.08e-3) -	3.5632e-2 (1.94e-3)	2.5288e-2 (5.54e-3) -	1.5789e-2 (2.68e-3)
HYL3	3.1095e-1 (5.04e-2) -	2.5405e-1 (6.90e-3)	1.8173e-1 (9.66e-3) -	1.0050e-1 (1.06e-3)	2.2674e-1 (1.56e-2) -	1.6325e-1 (1.71e-3)	1.6893e-1 (7.50e-3) -	1.3944e-1 (4.50e-3)
HYL4	2.8227e-1 (2.62e-2) -	1.1997e-1 (2.67e-2)	1.9973e-1 (1.17e-2) -	1.5200e-1 (1.29e-2)	2.2810e-1 (1.31e-2) -	1.5396e-1 (8.32e-3)	2.0906e-1 (1.02e-2) -	1.6397e-1 (4.28e-3)
HYL5	9.5460e-2 (1.67e-2) -	7.8557e-2 (1.05e-4)	4.6416e-2 (3.51e-3) -	3.8369e-2 (2.40e-3)	4.7632e-2 (3.84e-3) -	3.6285e-2 (2.79e-3)	6.3281e-2 (1.04e-2) -	3.3544e-2 (1.82e-3)
+/-/=	0/5/0		0/5/0		0/4/1		0/5/0	

表 S28 4 个代表算法及对应变种算法在 HYL 上获得的 PSP 均值与标准差

测试问题	DN-NSGAII	DN-NSGAII-AID	MMEA-WI	MMEA-WI-AID	MMOEA-GPD	MMOEA-GPD-AID	MO_Ring_PSO_SCD	MO_Ring_PSO_SCD-AID
HYL1	2.6671e+0 (4.19e-1) -	3.6328e+0 (5.84e-1)	4.6296e+0 (1.83e-1) -	5.4078e+0 (6.07e-2)	5.3263e+0 (4.71e-2) -	5.7024e+0 (3.30e-2)	4.0629e+0 (1.44e-1) -	4.6743e+0 (1.12e-1)
HYL2	5.0036e-1 (1.44e-1) -	1.4307e+0 (1.51e-1)	1.5613e+0 (3.20e-2) -	1.7057e+0 (2.16e-2)	1.7655e+0 (1.36e-2) -	1.9921e+0 (1.81e-2)	1.0279e+0 (1.68e-1) -	1.5715e+0 (3.07e-2)
HYL3	8.0231e-1 (1.23e-1) -	1.2222e+0 (2.51e-1)	1.1592e+0 (1.98e-2) -	1.6255e+0 (1.58e-2)	1.1821e+0 (2.23e-2) -	1.3285e+0 (2.94e-2)	9.3038e-1 (1.07e-1) -	1.1010e+0 (2.60e-2)
HYL4	2.7457e-1 (4.33e-2) -	1.0712e+0 (6.35e-2)	9.6051e-1 (1.42e-2) -	1.0250e+0 (1.22e-2)	9.9511e-1 (1.07e-2) -	1.0758e+0 (1.25e-2)	5.4115e-1 (1.10e-1) -	9.0582e-1 (1.65e-2)
HYL5	2.0676e+0 (4.20e-1) -	3.5318e+0 (2.60e-1)	3.5638e+0 (7.38e-2) -	7.6968e+0 (6.59e-2)	4.0295e+0 (4.43e-2) -	5.7815e+0 (4.28e-2)	2.3228e+0 (2.57e-1) -	4.1034e+0 (9.52e-2)
+/-/=	0/5/0		0/5/0		0/5/0		0/5/0	

4 运行时间对比

4.1 CEC 2019 MMOPs上各算法的运行时间结果

表 S29 MMODE-AID 与 7 个对比算法在 CEC 2019 MMOPs 上的运行时间结果对比

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MMF1	1.5615e+0 (5.33e-1) +	9.2692e+0 (1.80e+0) -	5.0134e+1 (3.37e+0) -	6.1169e+0 (3.01e-1) -	5.0418e+0 (3.64e-1) -	1.0404e+1 (7.62e-1) -	1.4731e+1 (8.39e-1) -	2.3682e+0 (1.53e-1)
MMF2	1.0013e+0 (6.16e-2) +	8.7823e+0 (4.93e-1) -	5.8638e+1 (4.74e+0) -	4.7167e+0 (2.14e-1) -	2.8885e+0 (2.21e-1) -	6.9667e+0 (4.50e-1) -	1.0657e+1 (1.86e+0) -	2.1438e+0 (1.35e-1)
MMF3	1.0509e+0 (9.02e-2) +	8.5330e+0 (4.24e-1) -	5.1879e+1 (7.83e+0) -	4.7918e+0 (2.25e-1) -	2.9094e+0 (2.78e-1) -	7.2167e+0 (5.33e-1) -	1.0582e+1 (1.76e+0) -	2.1273e+0 (6.43e-2)
MMF4	9.5786e-1 (5.74e-2) +	8.3787e+0 (4.07e-1) -	7.1891e+1 (6.47e+0) -	6.8779e+0 (3.06e-1) -	6.5722e+0 (2.51e-1) -	1.2042e+1 (5.80e-1) -	2.4013e+1 (1.06e+0) -	2.3293e+0 (1.04e-1)
MMF5	9.4929e-1 (5.85e-2) +	8.4114e+0 (4.61e-1) -	6.8889e+1 (6.07e+0) -	5.7840e+0 (2.37e-1) -	4.6916e+0 (3.22e-1) -	9.9975e+0 (4.87e-1) -	1.5739e+1 (7.84e-1) -	2.1926e+0 (8.90e-2)
MMF6	9.8414e-1 (8.95e-2) +	8.3890e+0 (4.33e-1) -	7.0410e+1 (4.76e+0) -	5.9451e+0 (2.05e-1) -	4.9430e+0 (1.78e-1) -	1.0603e+1 (4.42e-1) -	1.7237e+1 (8.13e-1) -	2.2627e+0 (1.04e-1)
MMF7	9.5452e-1 (6.59e-2) +	8.1074e+0 (4.23e-1) -	6.7242e+1 (5.63e+0) -	6.6036e+0 (2.98e-1) -	6.3730e+0 (2.72e-1) -	1.2569e+1 (5.56e-1) -	2.3904e+1 (1.22e+0) -	2.2659e+0 (1.10e-1)
MMF8	9.7392e-1 (5.89e-2) +	8.4557e+0 (4.25e-1) -	5.3351e+1 (6.02e+0) -	5.6833e+0 (2.41e-1) -	4.8839e+0 (2.73e-1) -	8.7101e+0 (4.63e-1) -	1.8946e+1 (1.01e+0) -	2.1158e+0 (8.04e-2)
MMF9	9.3907e-1 (6.69e-2) +	8.2437e+0 (3.98e-1) -	5.8257e+1 (5.26e+0) -	8.5704e+0 (4.13e-1) -	1.0648e+1 (4.02e-1) -	1.4772e+1 (6.10e-1) -	3.4558e+1 (1.60e+0) -	2.4129e+0 (1.14e-1)
MMF10	9.1399e-1 (5.55e-2) +	8.2859e+0 (4.24e-1) -	4.0713e+1 (6.88e+0) -	7.0398e+0 (7.63e-1) -	7.1319e+0 (1.78e+0) -	1.1068e+1 (2.17e+0) -	3.9179e+1 (2.08e+0) -	2.5247e+0 (2.03e-1)
MMF11	9.1862e-1 (5.82e-2) +	8.3870e+0 (4.43e-1) -	5.1678e+1 (8.03e+0) -	8.3366e+0 (3.32e-1) -	1.1152e+1 (4.20e-1) -	1.3590e+1 (7.20e-1) -	1.3590e+1 (1.57e+0) -	2.7063e+0 (1.33e-1)
MMF12	9.2977e-1 (6.22e-2) +	8.4416e+0 (4.11e-1) -	3.2679e+1 (2.12e+0) -	6.1795e+0 (2.26e-1) -	6.5620e+0 (2.76e-1) -	9.6870e+0 (5.13e-1) -	2.2835e+1 (1.28e+0) -	2.2280e+0 (8.47e-2)
MMF13	1.2581e+1 (1.26e-1) +	1.2581e+1 (5.61e-1) -	1.2612e+2 (1.27e+1) -	1.6509e+1 (5.16e-1) -	2.7489e+1 (1.38e+0) -	3.3856e+1 (1.53e+0) -	7.4024e+1 (2.77e+0) -	6.1774e+0 (2.87e-1)
MMF14	1.6276e+0 (9.70e-2) +	1.2931e+1 (6.27e-1) -	2.5644e+2 (3.43e+1) -	5.6138e+1 (2.29e+0) -	7.7769e+1 (4.27e+0) -	7.9201e+1 (2.50e+0) -	1.9013e+2 (7.75e+0) -	8.2481e+0 (5.06e-1)
MMF15	2.1027e+0 (1.54e-1) +	1.7614e+1 (1.28e+0) =	1.6214e+2 (1.88e+1) -	6.4965e+1 (2.65e+0) -	7.5997e+1 (3.48e+0) -	9.3859e+1 (3.71e+0) -	2.2303e+2 (1.25e+1) -	1.7634e+1 (1.50e+0)
MMF1_z	1.3027e+0 (1.76e-1) +	1.2231e+1 (1.26e+0) -	4.0973e+1 (2.63e+0) -	8.3679e+0 (8.79e-1) -	4.7831e+0 (2.05e-1) -	1.4373e+1 (1.24e+0) -	2.1591e+1 (1.61e+0) -	2.2104e+0 (7.80e-2)
MMF1_e	1.4038e+0 (1.18e-1) +	1.2151e+1 (1.14e+0) -	2.6004e+1 (3.16e+0) -	6.5446e+0 (7.02e-1) -	3.1139e+0 (1.90e-1) -	9.6932e+0 (9.77e-1) -	1.5936e+1 (1.52e+0) -	2.0639e+0 (8.88e-2)
MMF14_a	1.4456e+0 (2.78e-1) +	1.1249e+1 (1.65e+0) -	1.6754e+2 (2.01e+1) -	3.0558e+1 (1.14e+0) -	4.1740e+1 (1.40e+0) -	5.2716e+1 (1.72e+0) -	9.5903e+1 (3.48e+0) -	6.4622e+0 (3.04e-1)
MMF15_a	2.2204e+0 (2.54e-1) +	1.7258e+1 (1.54e+0) -	1.3806e+2 (8.24e+0) -	3.3594e+1 (1.85e+0) -	3.7101e+1 (1.18e+0) -	6.3186e+1 (1.97e+0) -	1.2296e+2 (4.66e+0) -	6.0941e+0 (2.65e-1)
Omni_test	1.4253e+0 (1.15e-1) +	1.1732e+1 (6.81e-1) -	1.0209e+2 (9.81e+0) -	1.1339e+1 (4.85e-1) -	1.3469e+1 (5.31e-1) -	1.8347e+1 (1.03e+0) -	4.1161e+1 (2.46e+0) -	4.5734e+0 (1.55e-1)
SYM_RART_simple	9.2677e-1 (7.27e-2) +	8.6852e+0 (6.55e-1) -	3.7468e+1 (3.12e+0) -	5.8218e+0 (3.84e-1) -	5.2258e+0 (3.05e-1) -	8.6065e+0 (5.85e-1) -	1.7076e+1 (1.03e+0) -	2.1483e+0 (8.11e-2)
SYM_RART_rotated	1.5714e+0 (2.39e-1) +	1.2178e+1 (1.10e+0) -	3.8466e+1 (4.17e+0) -	7.2668e+0 (8.01e-1) -	4.0029e+0 (1.68e-1) -	1.0652e+1 (8.86e-1) -	2.2269e+1 (1.78e+0) -	2.1480e+0 (8.96e-2)
+/-/=	22/0/0	0/21/1	0/22/0	0/22/0	0/22/0	0/22/0	0/22/0	

4.2 MPolygon上各算法的运行时间结果

表 S30 MMODE-AID 与 7 个对比算法在 MPolygon 上的运行时间结果对比

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MPolygon1	1.0483e+0 (1.71e-1) +	1.1326e+1 (5.78e-1) -	1.0265e+1 (7.43e-1) -	8.2801e+0 (8.91e-1) -	4.8101e+1 (7.92e+0) -	2.2145e+1 (5.40e+0) -	6.5924e+0 (3.11e-1) -	4.2830e+0 (1.79e-1)
MPolygon2	1.1643e+0 (9.52e-2) +	8.0409e+0 (4.86e-1) -	4.6132e+0 (3.92e-1) =	8.4456e+0 (4.02e-1) -	1.8076e+1 (6.38e+0) -	1.3661e+1 (4.03e+0) -	5.9287e+0 (2.20e-1) -	4.5814e+0 (1.78e-1)
MPolygon3	1.1101e+0 (1.09e-1) +	1.2652e+1 (5.72e-1) -	1.2223e+1 (6.74e-1) -	8.3248e+0 (4.29e-1) -	6.3222e+1 (7.50e+0) -	2.6082e+1 (6.50e+0) -	7.1602e+0 (2.64e-1) -	4.4753e+0 (2.16e-1)
MPolygon4	1.1863e+0 (1.47e-1) +	8.9859e+0 (8.25e-1) -	5.1913e+0 (5.18e-1) -	8.9905e+0 (8.28e-1) -	1.4179e+1 (7.92e+0) -	1.3897e+1 (4.44e+0) -	5.7278e+0 (3.33e-1) -	4.3804e+0 (2.06e-1)
MPolygon5	1.0276e+0 (4.43e-2) +	1.2958e+1 (7.64e-1) -	1.2571e+1 (7.72e-1) -	8.1749e+0 (3.38e-1) -	6.3331e+1 (6.69e+0) -	2.5194e+1 (6.19e+0) -	7.1403e+0 (3.09e-1) -	4.3490e+0 (1.73e-1)

表 S30 MMODE-AID 与 7 个对比算法在 MPolygon 上的运行时间结果对比 (续)

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
MPolygon6	1.0755e+0 (8.61e-2) +	8.8458e+0 (7.42e-1) -	4.9791e+0 (4.23e-1) -	8.4697e+0 (3.34e-1) -	1.6729e+1 (6.42e+0) -	1.3108e+1 (3.58e+0) -	5.7686e+0 (2.18e-1) -	4.3670e+0 (1.99e-1)
MPolygon7	1.0011e+0 (6.25e-2) +	1.3173e+1 (4.58e-1) -	1.3137e+1 (6.30e-1) -	8.3487e+0 (3.62e-1) -	6.7138e+1 (6.38e+0) -	2.8398e+1 (6.70e+0) -	7.3234e+0 (3.58e-1) -	4.4666e+0 (1.96e-1)
MPolygon8	1.0565e+0 (7.62e-2) +	8.8456e+0 (6.28e-1) -	5.2063e+0 (3.76e-1) -	8.7485e+0 (3.89e-1) -	1.2481e+1 (7.43e+0) -	1.3406e+1 (3.35e+0) -	5.8829e+0 (2.51e-1) -	4.3433e+0 (1.81e-1)
MPolygon9	1.0054e+0 (6.44e-2) +	1.3399e+1 (6.52e-1) -	1.3340e+1 (8.02e-1) -	8.4710e+0 (3.74e-1) -	6.6322e+1 (9.04e+0) -	2.9255e+1 (5.65e+0) -	7.2336e+0 (2.58e-1) -	4.4395e+0 (1.69e-1)
MPolygon10	1.0562e+0 (6.86e-2) +	9.1601e+0 (7.45e-1) -	5.3587e+0 (4.87e-1) -	8.7115e+0 (3.61e-1) -	1.3517e+1 (8.01e+0) -	1.4297e+1 (4.31e+0) -	5.8415e+0 (2.12e-1) -	4.3427e+0 (1.56e-1)
MPolygon11	1.0129e+0 (6.80e-2) +	1.3696e+1 (6.10e-1) -	1.3456e+1 (7.60e-1) -	8.4330e+0 (3.36e-1) -	6.7037e+1 (6.37e+0) -	3.0826e+1 (4.94e+0) -	7.3264e+0 (3.09e-1) -	4.4647e+0 (1.88e-1)
MPolygon12	1.0413e+0 (6.78e-2) +	9.4030e+0 (6.91e-1) -	5.2116e+0 (5.33e-1) -	8.7968e+0 (3.07e-1) -	1.3581e+1 (7.52e+0) -	1.5751e+1 (4.11e+0) -	5.8018e+0 (2.33e-1) -	4.2857e+0 (1.80e-1)
+/-/=	12/0/0	0/12/0	0/11/1	0/12/0	0/12/0	0/12/0	0/12/0	

4.3 HYL上各算法的运行时间结果

表 S31 MMODE-AID 与 7 个对比算法在 HYL 上的运行时间结果对比

测试问题	DN- NSGAI	MMEA-WI	MMOEA- GPD	MO_Ring_ PSO_SCD	MMODE_S PDN	CEA-LES	CMMO	MMODE-AID
HYL1	4.3821e+0 (8.56e-1) +	4.3151e+1 (1.76e+0) -	2.8200e+1 (1.57e+0) -	3.5902e+1 (1.73e+0) -	1.3974e+2 (4.17e+0) -	9.2423e+1 (4.29e+0) -	2.6183e+1 (8.74e-1) -	1.4783e+1 (6.33e-1)
HYL2	3.5581e+0 (1.83e-1) +	4.8373e+1 (2.06e+0) -	2.3706e+1 (2.03e+0) -	3.5878e+1 (2.28e+0) -	1.4731e+2 (6.56e+0) -	1.6083e+2 (9.06e+0) -	4.3106e+1 (2.29e+0) -	1.1375e+1 (4.52e-1)
HYL3	4.3763e+0 (2.39e-1) +	5.3338e+1 (2.79e+0) -	1.6025e+1 (1.94e+0) -	3.7035e+1 (2.34e+0) -	1.5729e+2 (5.06e+0) -	1.0113e+2 (5.65e+0) -	2.7189e+1 (1.16e+0) -	1.0301e+1 (4.03e-1)
HYL4	6.9467e+0 (2.59e-1) +	1.7248e+2 (4.64e+0) -	5.8136e+1 (1.03e+0) -	5.6266e+1 (1.87e+0) -	1.7601e+2 (4.74e+0) -	1.2075e+2 (1.76e+1) -	1.2618e+2 (4.26e+0) -	2.2633e+1 (7.18e-1)
HYL5	3.8502e+0 (1.79e-1) +	3.9667e+1 (7.26e-1) -	1.3819e+1 (1.48e+0) -	3.2496e+1 (1.10e+0) -	5.1379e+1 (3.25e+0) -	4.9670e+1 (2.96e+0) -	2.5290e+1 (7.66e-1) -	8.6329e+0 (3.56e-1)
+/-/=	5/0/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	0/5/0	

5 各对比算法在个体拥挤度和个体多样性计算方面的不同及差异化优势总结

表 S32 MMODE-AID 与 7 个对比算法在个体拥挤度和个体多样性计算方面的不同及各自差异化优势

算法	个体拥挤度计算	个体多样性计算	差异化优势
DN-NSGAI	基于拥挤距离 (Crowding Distance, CD)	仅直接采用决策 空间的 CD 值	继承 NSGA-II 的经典 CD 计算逻辑, 在低维、PS 数量少且无重叠的简单 MMOPs 中具备较好的计算效率。但因未结合目标空间信息, 对 PS 重叠或 PS 数量较多的问题, 易丢失决策空间多样性, 难以找到多个等效 PSs。
MMEA-WI	基于双 k 近邻 (Double k NN, DKN)	仅直接采用决策 空间的 DKN 值	对于决策空间 PS 为规则线段且无高维交互的 MMOPs, 对 PS 的识别与聚焦能力优于基于 CD 类算法。但因多样性计算仅局限于决策空间, 对于目标空间 PF 呈非凸形态的问题和超多目标问题, 无法有效平衡目标空间分布, 易出现 PF 覆盖不完整情形。
MMOEA-GPD	基于调和平均距 离 (Harmonic Average Distance, HAD)	仅直接采用决策 空间的 HAD 值	对于决策空间 PS 分布稀疏且分离明显的 MMOPs, 能有效保留稀疏区域的个体, 避免数学平均距离“拉平疏密差异”现象。但因仅依赖决策空间衡量多样性, 对于 PSs 重叠或者存在无限个 PSs 的问题, 无法动态判断决策空间与目标空间的映射关联, 难以同时维持两个空间的多样性。

MO_Ring_P SO_SCD	基于拥挤距离 (Crowding Distance, CD)	基于个体在决策 空间和目标空间 的 CD 最值	对于低维、PF 为凸且 PS 数量较少的 MMOPs, 能一定程度上平衡“决策空间覆盖”与“目标空间逼近”。但对于包含多个 PSs 的问题和超多目标问题, 决策空间与目标空间多样性平衡能力失效, 易出现找不全 PS 的情形。
MMODE_SP DN	基于聚类的特殊 拥挤距离 (Clustering- based Special Crowding Distance, CSCD)	基于个体在决策 空间和目标空间 的 CSCD 最值	对于 PS 为规则聚类分布的 MMOPs, 能识别不同 PS 簇并维持簇内多样性。但聚类依赖预定义参数, 对于不规则 PS 和无限 PS 问题, 易出现“错分/漏分”, 导致多样性计算失真。
CEA-LES	基于拥挤距离 (Crowding Distance, CD)	基于个体在决策 空间和目标空间 的 CD 最值, 再 结合 DKN 方式	对于中等复杂度、PS 数量适中且 PF 凸凹混合的 MMOPs, 能兼顾决策空间和目标空间的多样性平衡。但对于 PS 极多的问题和超多目标问题, 计算复杂度激增, 且决策空间和目标空间权重无自适应调整能力, 多样性维持稳定性不足。
CMMO	基于 k 近邻的密 度估计 (k NN Density Estimation, k DE)	基于个体在决策 空间和目标空间 的 k DE 值, 并 以固定权重方式 相加	对于 PS 为规则形状且决策空间和目标空间维度一致的 MMOPs, 可快速平衡决策空间和目标空间。但对于包含凹形 PF 等复杂问题和超多目标问题, 适应性较差。
MMODE- AID	基于自定义最近 邻平均距离 (Individual Crowdedness based on Nearest Neighbor averaged distance, ICNN)	自适应基于个体 在决策空间和目 标空间的整体拥 挤度 (Adaptive Individual Diversity based on global crowdedness of decision space and objective space, AID)	ICNN 可降低对边界的敏感程度, 且不需设置特定参数, 从而避免出现拥挤错觉, AID 可根据问题特性动态调整决策空间和目标空间的占比权重, 从而可对各类 PS 形状的密度分布进行合理判断, 且适配复杂 PS 场景, 总体具有更优的场景通用性与求解鲁棒性。不足之处在于计算逻辑相对更复杂带来的效率权衡。